

Multimodal Conversational AI for Mental Health Support: Integrating CV, NLP, and TTS in Real-Time.

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Abstract

This paper presents Kenko, an emotion-aware AI mental health assistant that integrates multiple deep learning models to provide personalized therapeutic support. The system combines natural language processing through a fine-tuned Llama-3 model with real-time computer vision for facial emotion recognition using DeepFace, and neural text-to-speech synthesis via Kokoro TTS. Kenko analyzes users' emotional states during conversations and adapts its therapeutic responses accordingly, creating a more empathetic and context-aware interaction. The system achieves real-time performance on consumer-grade GPUs and demonstrates the feasibility of multimodal AI systems for mental health support. We discuss the architecture, implementation challenges including memory optimization across multiple models, and the ethical considerations of AI-assisted therapy. An optional threat detection module using RF-DETR is explored for safety monitoring. Our results show that integrating emotional context significantly enhances the perceived empathy and relevance of AI-generated therapeutic responses, though we emphasize that such systems are supplementary tools rather than replacements for professional mental health care.

Introduction

Mental health challenges have reached epidemic proportions globally, with the World Health Organization estimating that one in four people will experience a mental health condition at some point in their lives. Despite this prevalence, access to professional mental health care remains severely limited. Barriers including cost, geographical constraints, long wait times, and persistent social stigma prevent millions from receiving timely support. The global shortage of mental health professionals is

estimated at 1.18 million practitioners needed to meet current demand which has created an urgent need for scalable, accessible supplementary interventions.

Artificial intelligence-based conversational agents have emerged as a promising approach to bridge this gap. Systems like Woebot [1], Wysa [2], and Replika [3] demonstrate that rule-based chatbots and retrieval-augmented dialogue systems can provide emotional support and cognitive behavioral therapy techniques at scale. However, these systems share a critical limitation: they operate exclusively through text-based interaction and lack awareness of users' non-verbal emotional cues. A user typing "I'm fine" while visibly distressed receives the same response as one who genuinely appears content. This emotional blindness fundamentally limits the therapeutic alliance since the relationship between therapist and client widely recognized as crucial for treatment efficacy.

Recent advances in computer vision and natural language processing present an opportunity to transcend these limitations. Deep learning models for facial emotion recognition, such as DeepFace [4], achieve human-level accuracy in detecting affective states from visual data. Simultaneously, large language models fine-tuned on therapeutic dialogue demonstrate remarkable capacity for empathetic, contextually appropriate responses. Neural text-to-speech systems like Dia TTS [5] now produce naturalistic voice synthesis indistinguishable from human speech. Yet despite these technological capabilities existing independently, their integration into a unified, real-time multimodal system for mental health support remains largely unexplored.

We present Kenko, an emotion-aware AI mental health assistant that synthesizes these advances into a cohesive therapeutic platform. Unlike existing text-only systems, Kenko continuously analyzes users' facial expressions via webcam during conversations, detects their emotional state in real-time, and dynamically adapts its therapeutic responses based on this visual context. The system combines four integrated components: real-time facial emotion recognition using DeepFace, a fine-tuned Llama-3 language model that incorporates emotional context into therapeutic dialogue generation, neural voice synthesis via Kokoro TTS (Previous attempts used Dia TTS) for natural spoken interaction, and an optional safety monitoring module using RF-DETR [6] object detection to identify potential self-harm implements. This multimodal architecture enables Kenko to respond not just to what users say, but to how they appear to feel which is a capability that more closely mirrors human therapeutic interaction.

The development of Kenko addresses several significant technical challenges. Running multiple large-scale deep learning models (totaling over 10 billion parameters) simultaneously requires careful memory optimization and architectural decisions. We implement 8-bit quantization for the language model, asynchronous processing for vision modules, and efficient state management to achieve real-time performance on

consumer-grade GPUs. The emotion detection module must balance responsiveness (updating frequently enough to capture mood shifts) with stability (avoiding oversensitivity to momentary facial movements). The language generation pipeline requires novel prompt engineering to seamlessly incorporate emotional context without disrupting conversational flow. Finally, integrating neural TTS which outputs discrete audio codes rather than raw waveforms that necessitates understanding and implementing proper decoding procedures rarely documented in deployment contexts.

Beyond technical contributions, Kenko raises important questions about the role of AI in mental health care. We do not position this system as a replacement for human therapists, but rather as a supplementary tool that can provide immediate, judgment-free support during moments of distress, between therapy sessions, or for individuals unable to access professional care. The system includes explicit disclaimers about its limitations and prompts users to seek professional help for crisis situations. The optional threat detection module represents an early exploration of safety considerations in AI mental health applications for how can systems identify and respond appropriately to signs of self-harm risk?

Looking forward, Kenko's modular architecture is designed with physical embodiment in mind. The current web-based implementation serves as a foundation for future integration with social robots, where multimodal perception (vision + language) and natural voice interaction could be augmented with physical presence, gestures, and mobility. Such embodied therapeutic agents could enhance the sense of companionship and reduce isolation, particularly for elderly populations or individuals in long-term care facilities. (As stated before, this system does not intend to replace real therapy. However, we do not discard the potential of these systems becoming “Intervention systems” that can act in the physical world through robotics, and help stop someone from committing self-harm.)

The key contributions of this work are:

- A novel multimodal architecture for emotion-aware AI therapy that combines real-time facial emotion recognition with context-adaptive language generation and neural voice synthesis
- Practical deployment solutions demonstrating how to run multiple large-scale deep learning models simultaneously on consumer GPUs (15~30GB VRAM) through quantization, asynchronous processing, and memory management techniques
- Prompt engineering methodology for incorporating visual emotional context into therapeutic dialogue without disrupting conversational coherence
- An exploration of safety considerations in mental health AI, including threat detection and appropriate system responses to crisis indicators

- An open-source implementation built with Gradio, providing a reproducible foundation for researchers and practitioners to build upon

The remainder of this paper is organized as follows: Section 2 reviews related work in AI-based mental health interventions, emotion recognition systems, and multimodal dialogue agents. Section 3 details Kenko's system architecture, including the design rationale for each component and their integration. Section 4 discusses implementation challenges and our solutions for real-time multimodal processing. Section 5 presents qualitative results, performance benchmarks, and preliminary user feedback. Section 6 addresses ethical considerations including privacy, transparency, and the appropriate scope of AI mental health tools and controversial potential enhancements. Section 7 concludes with limitations of the current system and directions for future work, including robotic embodiment and longitudinal evaluation studies.

Related Work (Background & Literature Review)

1.1 AI-Based Mental Health Interventions

The application of artificial intelligence to mental health support has evolved significantly over six decades, from early rule-based systems to contemporary neural language models. These systems aim to address the substantial gap between mental health care demand and professional availability, offering scalable, accessible interventions at reduced cost [7]. However, most existing solutions remain limited to text-based interaction and lack the capacity to perceive or respond to users' non-verbal emotional cues is a critical component of effective therapeutic communication.

1.1.1 Early Rule-Based Systems: ELIZA

The foundational work in conversational AI for mental health is Joseph Weizenbaum's ELIZA (1966) [8], developed at MIT to enable natural language conversation between humans and computers. Operating as a Rogerian therapist, ELIZA analyzed input sentences using decomposition rules triggered by keywords, then generated responses through pattern-matching reassembly rules [8]. Despite its technical simplicity and no semantic understanding or learning, ELIZA created surprisingly compelling interactions, with users anthropomorphizing the system and attributing genuine empathy to its responses [9].

Recent reinterpretations highlight ELIZA's historical mischaracterization. Shrager (2024) [10] argues ELIZA was not intended as a therapeutic chatbot but as a research platform for studying human interpretation and misinterpretation in human-machine communication. The program's fame stemmed from its fortuitous timing and unintended public adoption, leading to decades of misunderstanding about its original

purpose. Weizenbaum himself became an outspoken critic of "delusional thinking" about machine intelligence, warning against assumptions that human thought could be entirely computational [9].

Dillon (2020) [11] provides a gender critique of the "Eliza effect" which is the tendency to attribute human-like qualities to machines and tracing the nomenclature to Shaw's Pygmalion and exposing gendered assumptions embedded in perceptions of machine capability. This critique remains relevant as contemporary mental health AI systems often employ female-coded personas and voices.

Key limitations: ELIZA operated purely through pattern matching with no semantic understanding, emotional awareness, or learning capability. Its responses were deterministic templates, incapable of adapting to individual users or contexts beyond keyword matching. It represents the "illusion of understanding" rather than genuine therapeutic capability.

1.1.2 Contemporary Commercial Systems

Woebot

Woebot Health's conversational agent represents a significant evolution from rule-based systems, incorporating cognitive behavioral therapy (CBT) techniques into a mobile application format. The system engages users in structured therapeutic exercises through text-based conversation, tracking mood patterns and delivering evidence-based interventions [12].

Empirical evidence for Woebot's efficacy is mixed but generally positive. An 8-week study with 256 adults demonstrated significant reductions in perceived stress and burnout, with increased resilience across both clinical and subclinical populations [13]. Those with clinically elevated baseline symptoms showed greater improvements, suggesting the intervention scales to symptom severity. An adapted version for substance use disorders (W-SUDs) showed feasibility and significant improvements in cravings, confidence, and mood symptoms over 8 weeks, though study attrition was high (38%) [13].

However, a recent 18-week study with master's students found no significant anxiety or depression reduction and reported declining user acceptance due to technical glitches and language barriers [14]. This suggests Woebot's effectiveness may be context- and population-dependent, with usability barriers limiting real-world impact.

Technical approach: Woebot employs a hybrid architecture combining rule-based conversation flows with limited natural language understanding. The system follows scripted therapeutic protocols (primarily CBT) while incorporating basic sentiment analysis to adapt tone. It does not utilize large language models for open-ended dialogue generation.

Key limitations: Text-only interaction, scripted conversation paths limiting naturalistic dialogue, no visual emotion recognition, high attrition in some populations, and language/cultural limitations.

Wysa

Wysa is an AI-driven mental health app offering conversational support, mood tracking, and guided therapeutic exercises. Unlike Woebot's CBT-centric approach, Wysa incorporates a broader range of techniques including behavioral activation, mindfulness, and sleep interventions [15].

Evidence for Wysa's effectiveness is robust across diverse populations. Among 527 healthcare workers in Singapore during the COVID-19 pandemic, 80.1% completed at least two sessions (mean: 10.9 sessions over 3.8 weeks) [16]. The app showed particular engagement for anxiety and sleep interventions. For individuals with chronic musculoskeletal pain, Wysa engagement correlated with improved mental and physical health outcomes, with behavioral activation mediating anxiety reduction (indirect effect: -0.4, 80% CI: -0.7 to -0.1) [17].

A Singapore-based mixed-methods study reported that 91.6% of users successfully reframed negative thoughts through cognitive restructuring exercises, with single interactions significantly associated with future thought restructuring capability ($P < 0.001$) [16]. User satisfaction was high, with 83.03% rating helpfulness $\geq 3/5$.

Technical approach: Wysa uses a conversational AI engine combining natural language processing with rule-based therapeutic protocols. The system personalizes interventions based on user input patterns and self-reported symptoms but does not employ computer vision or physiological monitoring.

Key limitations: Similar to Woebot—text-only modality, no emotion recognition beyond text sentiment, scripted intervention modules, and dependence on user self-reporting of emotional states.

1.1.3 The Multimodal Gap

While Woebot and Wysa demonstrate clinical utility, they share a fundamental constraint: complete reliance on explicit user input. These systems cannot detect:

Facial expressions indicating incongruence between stated and felt emotions

Vocal prosody signaling distress despite reassuring words

Environmental context (e.g., presence of self-harm implements)

Physiological indicators of anxiety, depression, or suicidal ideation

This "input bottleneck" creates several problems:

Alexithymia barrier: Users who struggle to identify or articulate emotions receive limited benefit.

Social desirability bias: Users may minimize distress to avoid judgment or intervention.

Missed crisis signals: Systems cannot detect discrepancies between verbal reports and non-verbal cues that human therapists routinely observe.

Limited therapeutic alliance: The absence of perceived "seeing" undermines the sense of being understood.

Human therapists integrate multiple information channels like verbal content, facial expressions, body language, vocal tone, and contextual cues to assess emotional state and tailor interventions. Existing AI systems, by contrast, operate with only explicit textual input, fundamentally limiting their therapeutic sophistication.

1.1.4 Emerging Directions: LLM-Based Systems

Recent large language models (LLMs) like GPT-4, Claude, and Llama-3 demonstrate unprecedented natural language understanding and generation capabilities. Several research groups have explored their application to mental health:

ChatCounselor (Liu et al., 2023) [18]: Fine-tuned LLaMA for mental health support, demonstrating improved empathy and helpfulness over base models.

MentaLLaMA (Yang et al., 2024) [19]: Interpretable Mental Health Analysis on Social Media with Large Language.

GPT-5 Medical Benchmark [20]: OpenAI's GPT-5 Touts Medical Benchmarks and Mental Health Guidelines.

These LLM-based systems offer more naturalistic, open-ended dialogue than rule-based predecessors. However, they remain text-only and exhibit new failure modes including:

Overconfident responses on topics requiring professional judgment

Potential reinforcement of catastrophic thinking if not carefully guided

Lack of crisis detection capabilities

No integration with emotional or environmental sensing

The gap Kenko addresses: No existing system, whether rule-based (ELIZA, Woebot, Wysa) or LLM-based (ChatGPT variants) incorporates real-time visual emotion recognition to continuously adapt therapeutic responses based on observed affective state. This represents a fundamental limitation in the current landscape of AI mental health interventions.

Section 1.2: Emotion Recognition Systems

1.2.1 Facial Expression Recognition

Automatic facial expression recognition (FER) has seen a paradigm shift from traditional geometric, handcrafted feature-based methods to deep learning approaches that achieve remarkable accuracy, often rivalling human performance. Modern systems predominantly use convolutional neural networks (CNNs), which automatically learn robust feature representations from large labelled emotion datasets like FER-2013, AffectNet, and RAF-DB.

DeepFace: A unified platform for multimodal efficiency

The open-source DeepFace library is not a single model but a unified, model-agnostic interface that serves as a powerful framework for multimodal inference.

- **Model consolidation:** It supports multiple state-of-the-art face recognition and facial attribute analysis backends, such as VGG-Face, FaceNet, OpenFace, and ArcFace. This modular architecture allows developers to easily switch between models at runtime without having to rewrite their application code. Benchmarks often show FaceNet and ArcFace models outperforming the original Facebook DeepFace and human accuracy on datasets like LFW.
- **Decoupled pre-processing pipeline:** DeepFace handles the entire facial recognition pipeline, from detection and alignment to representation and verification. It encapsulates robust pre-processing steps that manage variations in illumination, head pose, and occlusions, ensuring that a normalized face is presented to the recognition model. This decoupled approach standardizes inputs, making it easier to integrate with other data sources.
- **Vector embeddings for multimodal fusion:** A key feature for multimodal efficiency is the library's ability to represent a face as a fixed-size vector embedding. This dense, numerical representation can be easily concatenated or fused with embeddings from other modalities, such as audio, text, or physiological signals. For instance, the facial emotion vector from DeepFace could be combined with a tone-of-voice vector from an audio model to create a richer, multimodal representation for tasks like explainable emotion evaluation.

Performance vs. research claims

The commonly seen 97.35% verification accuracy on the Labeled Faces in the Wild (LFW) dataset was a landmark achievement published by Facebook Research in 2014, not an inherent performance of the modern open-source DeepFace library itself. The modern library's performance depends on the chosen backend model; for example, benchmarks show FaceNet reaching over 99% on LFW, while the original Facebook model, when re-implemented, might yield lower scores due to different training or implementation details.

Alternative frameworks

Several alternative frameworks provide specialized features for facial analysis:

- **FER+:** Improves upon the standard FER-2013 dataset by relabeling ambiguous images with ten crowdsourced annotations, providing soft labels instead of hard ones. This accounts for human annotator disagreement and enables the training of more robust multi-label or statistical distribution-based FER algorithms.
- **MediaPipe Face:** A Google framework optimized for real-time performance on resource-constrained devices like mobile phones. It provides ultra-fast face detection and facial landmark tracking (using MobileNetV2) and outputs blendshape scores for detailed facial expression inference.
- **OpenFace:** An open-source toolkit focused on facial behavior analysis, offering real-time capability for webcam feeds. It is well-regarded for its fine-grained affective computing features, including Action Unit (AU) detection, gaze tracking, and head pose estimation. The upcoming OpenFace 3.0 promises a lightweight, multitask model with improved speed and accuracy.

For Kenko, we selected DeepFace for its balance of accuracy, ease of integration, and mature ecosystem. While MediaPipe offers superior speed, DeepFace's accuracy was prioritized for therapeutic contexts where misreading emotions could undermine user trust. OpenFace's AU-level granularity, while valuable for research, introduces complexity unnecessary for our application.

1.2.2 Multimodal Emotion Detection

Beyond facial expressions, comprehensive emotion recognition integrates multiple modalities:

Voice/Prosody: Pitch, tempo, energy variations signal emotional state

Text/Language: Sentiment analysis, linguistic patterns

Physiological: Heart rate variability, skin conductance, EEE

Body Language: Posture, gestures, movement dynamics

Multimodal fusion improves robustness, as single modalities may be ambiguous (e.g., neutral face with distressed voice). Early fusion combines raw features, intermediate fusion operates on learned representations, and late fusion aggregates modality-specific predictions. Recent work explores attention-based fusion and transformer architectures for temporal alignment.

Kenko currently employs unimodal visual emotion detection, prioritizing deployment simplicity and privacy (no audio recording beyond TTS output). Future work will explore voice prosody analysis during user input to create a bimodal affect recognition system.

1.2.3 Real-Time vs. Batch Processing

Emotion recognition systems face a fundamental trade-off between latency and accuracy. Batch processing allows sophisticated models and post-processing (temporal smoothing, multi-frame aggregation) but introduces unacceptable delays for interactive applications. Real-time systems require inference $< 100\text{ms}$ per frame to maintain conversational flow.

Kenko operates at 1-2 FPS for emotion updates, a design choice balancing:

Computational efficiency: Avoiding continuous frame processing reduces GPU load

Affective stability: Emotions persist across seconds, not frames

Context expiration: 60-second staleness threshold prevents outdated assumptions

This stands in contrast to systems like Affectiva (real-time dashboard) or EmoNet (video-level batch analysis), positioning Kenko between high-frequency monitoring and static assessment.

Section 1.3: Multimodal Dialogue Systems

1.3.1 Vision + Language Models

The integration of visual and linguistic modalities has progressed from early template-based image captioning to modern vision-language transformers (VLTs). Systems like CLIP, ALIGN, and Flamingo learn joint embeddings enabling zero-shot image-text understanding. GPT-4V and Gemini extend large language models with native visual input, supporting conversational interaction grounded in images.

However, these general-purpose VLMs differ fundamentally from Kenko's architecture:

Static vs. continuous: VLMs process individual images on-demand; Kenko maintains persistent emotion state across conversation

Explicit vs. implicit: Users must explicitly reference images ("What's in this photo?"); Kenko silently observes without user prompting

General vs. specialized: VLMs handle arbitrary visual tasks; Kenko focuses exclusively on affective understanding

Kenko's design pattern resembles context-aware assistants [33] that accumulate environmental state (location, time, activity) to inform responses, extending this paradigm to emotional context.

1.3.2 Embodied Conversational Agents

Embodied conversational agents (ECAs) combine dialogue capabilities with physical or virtual presence, often including gesture generation, gaze behavior, and facial animation. Historical systems like ELIZA-with-avatar and more sophisticated agents like Greta and SEMAINE demonstrated that embodiment enhances perceived social presence and engagement.

Social robots such as Pepper, NAO, and Furhat represent physically embodied ECAs. These systems typically employ scripted behaviors triggered by keyword detection or predefined interaction states. Recent work explores data-driven gesture generation and affective body language, but most deployed robots remain rule-based.

Kenko's current web-based interface lacks embodiment but is architected for future robotic deployment. The modular design (Section 3) separates perception (emotion/threat detection), cognition (LLM dialogue), and expression (TTS) to facilitate integration with robot middleware like ROS2.

1.3.3 Prior Work on Emotion-Aware Dialogue

Several systems have explored emotion-conditioned dialogue generation:

EmpatheticDialogues dataset: 25K conversations grounded in emotional situations, enabling training of empathy-aware models

MIME: Multi-modal empathetic dialogue system integrating text, audio, and visual cues (images, not video)

EmpBot: Retrieval-based system selecting responses matching detected user emotion

Emotional Support Conversation (ESC): Dataset and models for providing comfort during distress

These systems demonstrate emotion-aware NLG but with key differences from Kenko:

Dataset-based vs. real-time: Most use annotated emotional labels from text; Kenko infers affect from live video

Retrospective vs. continuous: MIME analyzes images users share; Kenko monitors ongoing facial expressions

Research vs. deployment: Most are academic prototypes; Kenko targets real-world therapeutic use

The closest prior work is EmoRL, which uses reinforcement learning to optimize empathetic responses based on simulated user emotional trajectories. However, EmoRL operates in text-only simulation without visual perception.

Kenko's novelty: We are unaware of prior systems combining (1) continuous visual emotion monitoring, (2) large language model dialogue generation with injected affective context, and (3) natural voice synthesis in a unified therapeutic application.

Section 1.4: Neural Text-to-Speech

1.4.1 Evolution from Concatenative to Neural TTS

Text-to-speech synthesis has progressed through three major paradigms:

Concatenative TTS (1980s-2000s): Stitches pre-recorded speech units (diphones, triphones) selected from large databases. Produces natural prosody for recorded phrases but struggles with novel contexts and sounds robotic at unit boundaries.

Statistical parametric TTS (2000s-2010s): Models acoustic features (spectrograms) using HMMs or DNNs, then synthesizes audio via vocoders (STRAIGHT, WORLD). Enables flexible voice modification but produces muffled, over-smoothed output.

Neural end-to-end TTS (2016-present): Directly predicts audio waveforms or high-resolution spectrograms from text using deep learning. Tacotron and Tacotron 2

pioneered sequence-to-sequence TTS with attention mechanisms, achieving human-like prosody. WaveNet introduced autoregressive waveform generation, dramatically improving naturalness despite slow inference.

MODEL	ARCHITECTURE	SAMPLE RATE	SPEED	VOICE CLONING	KEY INNOVATION
Tacotron 2	Seq2seq + WaveGlow	22 kHz	Slow	No	Attention based prosody
Fast Speech 2	Non-autoregressing	22 kHz	Fast	No	Parallel generation
VITS	VAE+adversarial	22 kHz	Fast	Limited	End-to-end stochastic
Your TTS	VITS variant	24 kHz	Fast	Yes (1-shot)	Zero-shot multilingual
XTTSv2	Transformer + code c	24 kHz	Medium	Yes (6s ref)	17 languages, streaming
Bark	GPT-style audio LM	24 kHz	Slow	Limited	Expressive, music, SFX
Parler-TTS	Conditional DETR	44.1 kHz	Slow	No	Natural Language prompts
Dia	Audio code c + LM	44.1 kHz	Slow	Yes (6s ref)	High fidelity, few-shots
Kokoro-TTS	Style TTS 2	24 kHz	Medium-Fast	No	High efficiency for size

Key trends:

Audio codecs: Models like Dia, XTTS, and Vall-E decompose speech into discrete tokens (codebooks), enabling language model-style generation. This trades raw waveform generation for compressed representations decoded by neural vocoders.

Voice cloning: Few-shot adaptation from short reference audio (3-10 seconds) enables personalized voices without full fine-tuning.

Multilingual: Single models supporting 10+ languages via learned phoneme representations like Kokoro-TTS.

1.4.2 Selection Rationale: Kokoro TTS

For Kenko, we selected Kokoro TTS based on several criteria:

Advantages:

- High audio quality: 24.0 kHz sampling rate, important for therapeutic contexts where voice quality impacts perceived empathy and professionalism.
- Reasonable latency: 5-15 seconds for 200-500 character responses on T4 GPU, acceptable for conversational interaction (humans tolerate 2-4s "thinking" pauses).
- Size: The model has a size of 82 million parameters which helps in making efficient inferences on consumer grade GPUs.
- Open-source deployment: No API costs or internet dependency, critical for privacy-sensitive mental health applications

Trade-offs:

- Cloning: Unlike models like Dia, Kokoro lacks native cloning abilities which can be an issue for later developments.
- Lack of explicit emotional control: The model generates natural intonation, but users cannot easily input parameters to explicitly control emotional styles like "happy," "sad," or "angry." The expressiveness is generated somewhat organically by the model's style diffusion system, making fine-grained control challenging compared to models designed with explicit emotion tags.

Alternatives considered:

- XTTS v2: Comparable quality at 24 kHz; we prioritized Dia's 44.1 kHz for audio fidelity.
- Bark: Superior expressiveness but prohibitive latency (15-30s); unsuitable for real-time dialogue.
- ElevenLabs API: Excellent quality but introduces API costs (\$0.30/1K chars), privacy concerns (cloud processing), and internet dependency.
- Dia represents the optimal quality, deployability, and privacy for therapeutic applications; however, its size of 1.6B increases the generation time drastically, almost twice the time of Kokoro-TTS

Section 1.5: Safety in AI Mental Health

1.5.1 Crisis Detection Systems

Existing AI systems employ keyword-based detection, sentiment thresholds, and machine learning classifiers trained on annotated crisis messages to identify suicidal ideation. However, these approaches have limitations:

High false positive rates (22%)[21] leading to alert fatigue

Cultural and linguistic biases in training data

Inability to detect non-verbal crisis signals (e.g., self-harm preparation)

Kenko's threat detection module represents a complementary approach: visual observation of environmental risk factors (weapons, medications). This shifts from predictive modeling (estimating future self-harm) to reactive monitoring (observing current risk indicators).

1.5.2 Ethical Frameworks for Intervention

The mental health AI community lacks consensus on intervention ethics. Most systems adhere to "do no harm through inaction", limiting responses to verbal support and crisis resource provision. However, embodied robots capable of physical action challenge this paradigm.

Sharkey & Sharkey (2012) discuss intervention in eldercare robots, arguing that physical capability creates moral obligation. Coeckelbergh (2010) proposes a capabilities approach, suggesting robots should enable human flourishing while respecting autonomy. These frameworks inform our tiered intervention proposal (Section 6.5).

Section 2: System Architecture

2.1: Overview

Kenko operates as a **multimodal pipeline** integrating three primary subsystems and a central state management layer (Figure 1). The architecture is designed for modularity, allowing components to operate asynchronously while sharing contextual information through a unified state store. This design facilitates graceful degradation. If emotion detection fails, the dialogue system continues functioning with reduced context, and enables future extensions.

The three core subsystems are:

1. **Emotion Recognition Module:** Real-time facial recognition analysis via DeepFace
2. **Contextual Language Generation Module:** Emotion-aware therapeutic dialogue with advanced memory
3. **Speech Synthesis Module:** Natural voice output via Kokoro-TTS

These components communicate through a **shared state management system** that maintains temporal context across all modalities. Unlike traditional chatbots that process each query independently, Kenko maintains persistent emotional and conversational memory, enabling genuinely context-aware therapeutic interactions.

Main architecture principles:

1. **Modularity:** Each subsystem can be developed, tested, and optimized independently
2. **Asynchronous operation:** Emotion detection and dialogue generation run in parallel to minimize latency
3. **Graceful degradation:** Systems functions even if emotion detection fails (text-only mode)
4. **Privacy-first design:** All processing occurs locally; no data transmitted to external servers

2.2 Emotion Recognition Module

2.2.1 Architecture

- **Input:** 640x480 RGB webcam frames at 1 FPS
- **Model:** DeepFace with OpenCV backend
- **Output:** 7-Class emotion distribution (angry, disgust, fear, happy, sad, surprise, neutral)

2.2.2 Implementation Details

The emotion recognition module maintains a global state dictionary:

```
current_emotion_state = {  
    "dominant": "neutral",  
    "confidence": 0.0,  
    "all_emotions": {},  
    "last_update": None  
}
```


The module operates **asynchronously** at 1 FPS, as deliberate design choice balancing three constraints:

1. **Computational efficiency:** Processing every frame would consume excessive GPU resources needed for language generation
2. **Affective stability:** Human emotions persist across seconds, around 60-90 to be precise, not milliseconds [22]. Sampling at 1 Hz captures meaningful state changes without over-sensitivity to transient facial movements (e.g., blinking, talking)
3. **Context freshness:** A 60-second expiration threshold prevents stale emotional assumptions

We employ `enforce_detection=False` to maintain continuity when faces are temporarily obscured (looking away, adjusting position). This trades occasional false negatives for smoother user experience.

2.2.3 Emotion Context Injection

Detected emotions are injected into a language model's prompt as structured context:

[User's Current Detected Emotion: sad (73.2% confidence)]

This simple format provides the LLM with explicit affective state without requiring fine-tuning for emotion-conditioned generation. The confidence score enables the model to modulate response intensity, high confidence triggers stronger empathetic pivots, while low confidence maintain cautious neutrality.

Rationale: Real-time emotion detection allows Kenko to identify discrepancies between verbal content and non-verbal affect; a critical capability for therapeutic contexts. A user stating "I am fine" while displaying sadness triggers different response strategies than genuine contentment.

2.3 Advanced Memory System

2.3.1 Motivation

Traditional chatbots suffer from **context amnesia** [23] where each interaction is treated as isolated, with no accumulation of relationship history, recurring themes, or therapeutic progress. Human therapists, by contrast, maintain rich memory:

- **Episodic memory:** Specific past events ("You mentioned your mother's passing last month")

- **Semantic memory:** Generalized themes (“You often struggle with work-life balance”)
- **Emotional memory:** Affective patterns (“Discussing relationships seems to trigger anxiety”)

Kenko implements a **multi-tier memory architecture** inspired by human cognitive models [24], enabling longitudinal therapeutic relationships within the session.

2.3.2 Memory Tiers

Tier 1: Working Memory (Immediate Context)

- **Capacity:** Last 5 conversations exchanges (~10 turns)
- **Purpose:** Maintain immediate conversational coherence
- **Decay:** Oldest exchanges consolidated to short-term memory when capacity is full
- **Access:** Full verbatim recall

Tier 2: Short-Term Memory (Recent Session)

- **Capacity:** Last ~50 exchanges with temporary decay
- **Purpose:** Track session-level themes and emotional trajectory
- **Decay function:** Exponential decay based on time elapsed and importance score:

$$\text{Importance}(t) = \text{importance}_0 \times \exp(-\lambda \times \Delta t)$$

where $\lambda = 0.01$ (decay rate) and Δt = minutes elapsed

- **Importance scoring:** Calculate via multi-factor weighing:
 - Emotional intensity (± 0.3)
 - Message length/detail (± 0.2)
 - Therapeutic keywords (± 0.3)
 - Base importance: (0.5)

Tier 3: Long_Term Memory (Semantic Clusters)

- **Capacity:** Unlimited, organized into semantic clusters
- **Purpose:** Retain recurring themes, relational patterns, progress indicators
- **Storage:** Vector embeddings (384-dim via all-MiniLM-L6-v2) stored in ChromaDB
- **Clustering:** DBSCAN-inspired semantic clustering groups similar memories (cosine similarity threshold: 0.7)

- **Retrieval:** Clusted centroid enable thematic recall (“previous discussion about anxiety”)

Tier 4: Emotional Memory (Affective Patterns)

- **Tracks:**
 - Emotion transitions (e.g., sad > neutral after discussing coping strategies)
 - Trigger patterns (what contexts provoke which emotions)
 - Coping effectiveness (which interventions correlate with improved affect)
- **Use case:** Identify patterns like “User become anxious when discussing work; mindfulness exercises previously helped”

2.3.3 Memory Consolidation Process

Memory flows through tiers via **consolidation events**:

1. **Working > Short-term:** When working memory exceeds capacity (>5 exchanges), oldest entry consolidates with calculated importance score
2. **Short-term > Long-term:** When importance decays below threshold (0.15), memory clusters into semantic long-term storage
3. **Emotional pattern extraction:** Concurrent with consolidation, emotional transitions and trigger contexts are logged

This mimics human memory consolidation during sleep [25], where episodic memories are gradually abstracted into semantic knowledge.

2.3.4 Contextual Retrieval

When generating responses, Kenko retrieves relevant memories via **multi-tier hybrid retrieval**:

```
def retrieve_contextual_memory(self, query, current_emotion):
    """Advanced retrieval using multiple memory tiers"""
    context = {
        "working": [],
        "short_term": [],
        "long_term": [],
        "emotional": [],
        "themes": []
    }
```

}

Retrieval scoring combines:

- **Semantic similarity:** Cosine distance between query embedding and memory embeddings
- **Importance weighting:** Higher importance memories prioritized
- **Emotional congruence:** Memories matching current emotion receive boost (x1.2)
- **Recency:** More recent short-term memories slightly favored

Example prompt with context:

(Recent Conversation)

User: I had a panic attack at work yesterday

Kenko: That sounds frightening. Can you tell me what was happening?

(Important Recent Context)

User: My boss has been criticizing my performance

User: I've been having trouble sleeping for weeks

(Related Past Discussions)

User: I've always struggled with perfectionism

Kenko: Perfectionism can create immense pressure...

(Emotional Pattern)

Previously triggered by: work stress, performance reviews

(Session Themes)

Anxiety, work, sleep, perfectionism

(Recurring Concerns)

Anxiety, work stress

This rich contextual grounding enables responses that reference past discussions, identify patterns and track therapeutic progress, all capabilities absent in stateless chatbots.

3.3.5 User Model Construction

Beyond conversation memory, Kenko constructs a psychological profile tracking:

- **Communication style:** Detailed vs. Concise vs. Moderate (affects response length)
- **Recurring concerns:** Automatically detected from keyword patterns (anxiety, relationships, work stress, etc.)
- **Progress indicators:** Tracked via sentiment trends and coping strategy mentions
- **Relational patterns:** Emerging themes in interpersonal relationships

This user model enables **personalization** without explicit configuration, adapting therapeutic style to individual needs.

2.4 Contextual Language Generation Module

2.4.1 Architecture

- **Base Model:** Llama-3.2-3B fine-tuned on mental health dialogue (IniNLP247/Kenko-mental-health-llama-3-model)
- **Quantization:** 8-bit via BitsAndBytes (reduces memory from ~7GB to ~4GB), though it is important to note that the quantization version is mostly local since HF Spaces does not allow 8-bit.
- **Context Window:** 4096 tokens
- **Generation Parameters:**
 - Temperature: 0.7 (balanced creativity/consistency)
 - Top-p: 0.9 (nucleus sampling)
 - Repetition penalty: 1.1 (discourages loops)
 - Max new tokens: 512 (concise therapeutic response)

2.4.2 Prompt Engineering

Kenko constructs **dynamic prompts** incorporating context streams:

```
conversation = ""

for user_msg, bot_msg in history:
    conversation += f"User: {user_msg}\nKenko: {bot_msg}\n\n"

emotion_context = get_emotion_context()
```

```

prompt = f"""### Instruction:
You are Kenko, a compassionate mental health therapist. Provide
empathetic, helpful, and professional responses to support the user's
mental wellbeing.

{emotion_context}

{conversation}User: {message}

### Response:

```

Key innovation: Unlike static prompts, this structure dynamically integrates:

1. System role definition
2. Real-time visual emotion state
3. Multi-tier memory retrieval
4. Current user input

This enables the model to respond to **implicit** emotional cues (visual affect) and **longitudinal** context (memory) simultaneously.

2.4.3 Response Generation Pipeline

1. User submits message
2. Retrieve contextual memory based on message and current emotion
3. Construct prompt with emotion + memory context
4. Generate response via Llama-3 (4-7s on T4 GPU)
5. Post-process: Trimp, validate appropriateness
6. Update memory tiers with new exchange
7. Track emotional transition if emotion changed

Latency breakdown:

- Memory retrieval: ~0.5-1.0s
- Prompt construction: ~0.05s
- LLM inference ~3-5s
- Memory update: ~0.3s

Total: ~4.0-6.6s

2.5 Speech Synthesis Module

2.5.1 Architecture

- **Model:** Kokoro-TTS
- **Sample Rate:** 24.0 kHz (High audio quality)
- **Generation Pipeline:**
 - Text > Audio codes (discrete codebook indices)
 - Audio codes > Waveform (neural vocoder decoding)

2.5.2 Implementation Details

Kokoro-TTS deploys in a single stage using a unified pipeline:

```
def generate_tts(text):
    try:
        text = text[:600]
        generator = tts_pipeline(
            text, voice='af_heart',
            speed=1, split_pattern=r'\n+'
        )
        audio_chunks = []
        for gs, ps, audio in generator:
            audio_chunks.append(audio)
        if not audio_chunks:
            print("TTS generation failed")
            return None
        audio_array = np.concatenate(audio_chunks, axis=0)
        audio_array = audio_array.astype(np.float32)
        sample_rate = 24000
        return (sample_rate, audio_array)
    except Exception as e:
        print(f"TTS generation error: {str(e)}")
        return None
```

This unified pipeline function handles the entire process end-to-end seamlessly by:

- **Text Processing/Phonemization**
- **Acoustic Modeling (StyleTTS 2)**
- **Vocoding (ISTFTNet)**

2.5.3 Performance Characteristics

- **Latency:** 5-15 seconds for 200-600 character response on a T4 GPU
- **Real-time factor:** ~0.4 (generates 1s audio in 0.4s)
- **Quality:** High naturalness, decent articulation, appropriate prosody for therapeutic contexts

Trade-off analysis:

While better TTS models exist [eg:., Dia-TTS], Kokoro-TTS' 24.0 kHz output provides high-grade audio quality critical for therapeutic credibility. Preliminary user feedback indicates voice quality significantly impacts perceived empathy, and the performance has a staggering contrast to Dia

2.5.4 Autoplay Integration

Generated audio automatically plays via Gradio's Audio component with `autoplay=True`. This maintains conversational flow without requiring the user to click. Essentially mimicking natural spoken dialogue.

2.6 State Management & Synchronization

2.6.1 Global State Variables

Kenko maintains three global state objects:

```
current_emotion_state = {
    "dominant": "neutral",
    "confidence": 0.0,
    "all_emotions": {},
    "last_update": None
}

advanced_memory = AdvancedMemorySystem(embedding_model,
global_vector_store)

previous_emotion = "neutral"
```

2.6.2 Temporal Logic

- **Emotion staleness:** Context expires after 60 seconds

- **Memory decay:** Short-term memories decay exponentially ($\lambda=0.01/\text{min}$)
- **Consolidation triggers:** Capacity limits (working), importance threshold (short-term)

Asynchronous updates: Emotion detection runs independently of dialogue generation, updating global state accessed during prompt construction. This decoupling allows graceful degradation; if webcam fails, system continues with text-only interaction.

2.7 User Interface

2.7.1 Framework

- **Technology:** Gradio 4.x (Python-based web UI)
- **Deployment:** Web-based, accessible via browser
- **Streaming:** WebRTC for webcam, WebSocket for real-time updates

2.7.2 Layout

Left Column (2/3 width):

- Chat interface with user/assistant avatars
- Audio player (autoplay enabled)
- Text input with Enter-to-send
- Clear chat and example prompts buttons

Right Column (1/3 width):

- Live webcam feed
- Emotion distribution (bar chart, top 7 classes)
- Real-time status indicator

Interaction Flow:

1. User enables webcam > Emotion detection starts (1 FPS loop)
2. User types message > Click Send
3. System generates response (5-7s) + TTS (8-15s)
4. Text appears in chat, audio plays automatically
5. Loop continues with updated emotional + memory context

2.8 Deployment Architecture

2.8.1 Current: HuggingFace Spaces (ZeroGPU)

- **GPU:** NVIDIA H200 (142GB VRAM) via ZeroGPU allocation
- **Frameworks:** spaces.GPU decorator for on-demand GPU access
- **Public URL:** Persistent, shareable link
- **Constraints:** Cold start latency (~30s when idle), 60s execution timeout

Memory footprint (with FP16)

- Llama-3.2-3B (FP16): ~6.4GB
- Kokoro TTS: ~350MB
- DeepFace + embeddings: ~1GB
- ChromaDB: <500MB
- **Total:** ~8.3GB (fits perfectly on 141GB H200)

2.8.2 Alternative deployments

Local(Development):

- Google Colab with T4 GPU (15GB VRAM)
- Can run 8-bit quantization for testing and improvements

Production Options:

- **RTX A6000 (48GB):** Dedicated instance, no cold starts
- **Self-hosted:** In-premises deployment for institutional use (hospitals, universities)
- **Edge deployment:** Raspberry PI + Coral TPU for emotion detection, potential STT, cloud API generation (hybrid architecture)

2.9 Future: Robotic Embodiment

Kenko's modular architecture is designed for physical embodiment. The current system maps to robot components:

Kenko Module	Robot Analog	Integration
Emotion recognition	Face camera + CV	Already implements
Language generation	LLM inference's API	Already implemented
TTS	Robot speaker	Direct audio output
STT	Robot microphone	Direct audio input and translation to text with VOSK

2.10 Key Design Decisions

Decision	Rationale	Trade-off
1 FPS emotion detection	Balance responsiveness vs. efficiency	May miss brief micro-expressions
60s emotion expirations	Prevents stale assumptions	May lose context if user pauses
Multi-tier memory	Enables longitudinal relationships	Added complexity vs. stateless
8-bit quantization (local tests)	Fit models on consumer GPUs	Slight accuracy loss (~2-3%)
Autoplay TTS	Maintain conversational flow	Potential browser autoplay blocks
Gradio framework	Rapid prototyping, easy deployment	Less customizable than React/Vue
No local storage	Privacy first, no persistent tracking	Session memory lost on close

Section 3: System Architecture

3.1: Model Selection & Justification

3.1.1 Language Model: Llama-3.2-3B

We selected **Llama-3.2-3B** as the foundation for Kenko’s conversational capabilities based on several critical factors:

Advantages:

- 1. Open-source deployment:** Unlike GPT-5, Gemini, or Claude, Llama can be self hosted without API costs or cloud dependencies, critical for privacy-sensitive mental health applications
- 2. Instruction-tuned baseline:** The base Llama-3.2-3B-Instruct model follows conversational prompts reliably without task-specific fine-tuning.
- 3. Fine-tunability:** Open weights enable domain-specific adaptation on mental health dialogue datasets
- 4. Manageable size:** 3B parameters offer strong performance while remaining highly deployable on consumer/cloud GPUs (vs. +10B models requiring multi-GPU setups)

- 5. **Active ecosystem:** Extensive community support, quantization, tools (BitsAndBytes), and deployment frameworks (vLLM, TGI)

Fine-tuning Process:

The base Llama-3 model was fine-tuned on a curated mental health dialogue dataset to produce IniNLP247/Kenko-mental-health-llama-3-model. The training corpus included:

- Therapeutic conversation transcripts from another LLM
- Crisis counselling dialogues
- Empathetic responses

Training challenges:

- **Resource constraints:** Fine-tuning 3B parameters requires substantial compute (minimum 80GB GPU memory for full fine-tuning)
- **Solution:** Used QLoRA through the Axolotl framework within RunPod cloud services to train using an H100
- **Hyperparameters:** Learning rate $2e-5$, batch size 8, with gradient accumulation (effective batch 16), trained for 1.5 epochs
- **Duration:** 15 hours

Alternative models considered:

- **GPT-5:** Superior quality but prohibitive API costs and cloud dependency compromise privacy
- **Mistral-7B:** Comparable performance but Llama-3 has better instruction-following
- **Llama-4-Maverick-17B-Instructs:** Higher quality and MoE architecture, but requires a higher level of processing power and storage

3.1.2 Emotion Recognition: DeepFace

Selection rationale:

DeepFace was chosen over alternatives (MediaPipe, FER+) for several reasons:

1. **Accuracy:** ~97% on benchmarks like LFW, approaching human-level performance.
2. **Ease of integration:** Single-line API (DeepFace.analyze()) abstracts model complexity
3. **Backend flexibility:** Supports multiple face recognition models (VGG-Face, Facenet, OpenFace) switchable runtime
4. **Robust preprocessing:** Handles illumination, pose and occlusion variations better than lightweight alternatives

5. **Mature ecosystem:** Well-documented, actively maintained, extensive issue resolution history

Trade-offs accepted:

- **Speed vs. Accuracy:** DeepFace is considered better in accuracy compared to MediaPipe, not speed. It achieves **97–98% accuracy** on benchmarks like LFW, approaching human-level performance, due to its deep neural network trained on massive datasets. However, this comes at the cost of speed, processing only **2–3 FPS** and requiring significant computational resources (~500MB model size). In contrast, MediaPipe prioritizes speed (up to **60 FPS**) with lightweight processing, making it suitable for real-time applications, but it relies on simpler models or external classifiers for tasks like emotion detection, resulting in lower accuracy unless combined with other systems.
- **Granularity:** OpenFace provides Action Unit (AU) detection for fine-grained analysis. We deemed this unnecessary complexity for our system since the 7-class emotion taxonomy (Etkman’s universal emotions) suffice for therapeutic context injection.

Deployment configuration:

```
result = DeepFace.analyze(  
    img_path=image,  
    actions=['emotion'],  
    enforce_detection=False,  
    detector_backend='opencv'  
)
```

3.1.3 Text-to-Speech: Kokoro TTS

Selection rationale:

After evaluating multiple neural TTS systems, Dia-1.6B emerged as optimal:

Criterion	Parler TTS	XTTS v2	Dia	Kokoro (Selected)
Sample rate	44.1 kHz	24 kHz	44.1 kHz	24.0 kHz
Latency (500 chars)	500+ sec	10-15 sec	15-30 sec	10-20 sec
Voice cloning	No	Yes (6s ref)	Yes (6s ref)	No

Deployment	Complex	Moderate	Moderate	Simple
Audio quality	Medium	High	Excellent	High

Critical discovery: Audio codes vs. Waveforms

Initial implementations attempts with Parler TTS failed catastrophically at first with generations times exceeding 500 seconds on T4 GPU, later tests worked but the quality wasn't up to standard. Investigation revealed Parler's generation parameters were incompatible with real-time use. After testing XTTS, and Dia; we discovered Kokoro.

High speed model justification:

Though higher quality is a much better option for therapeutic reasons, the slow generation time of Dia proved to be a problem, so it was prioritized to focus on speed while quality is a second concern. Preliminary user feedback indicates voice quality significantly impacts perceived empathy and system credibility since a robotic or low-fidelity voice undermines therapeutic alliance. The additional compute cost though justifiable in some cases, has to be addressed, for this reason, Kokoro took Dia's place.

3.1.4 Memory & Embeddings

Embedding model: all-MiniLM-L6-v2

- **Rationale:** 384-dimensional sentence embeddings, optimized for semantic similarity tasks
- **Speed:** ~500 sentences/sec on CPU, enabling real-time memory retrieval
- **Size:** 80MB model, minimal memory footprint
- **Alternative considered:** all-mpnet-base-v2 (768-dim, higher quality) rejected due to 2x slower inference

Vector store: ChromaDB

- **Rationale:** Lightweight, embeddable vector database with no external server requirements
- **In-memory mode:** Entire vector store resides in RAM, enabling <100ms retrieval
- **Limitation:** No persistence across sessions

3.2 Deployment Strategy: Hugging Face Spaces with ZeroGPU

3.2.1 Infrastructure

Platform: Hugging Face Spaces (Pro tier with ZeroGPU access)

GPU allocation:

- **Hardware:** NVIDIA H200 (141GB HBM3e)
- **Allocation model:** On-demand via @spaces.GPU decorator
- **Concurrency:** Shared pool, allocated pre-request
- **Timeout:** 60-second execution limit pre request

Why ZeroGPU over dedicated instances:

- **Cost efficiency:** 9\$/month vs. \$1.50-6.90/hour dedicated H100/H200
- **Scalability:** Automatic scaling during high traffic
- **Simplicity:** No infrastructure management, automatic deployment from Git
- **H200 access:** 141GB VRAM eliminates memory constraints entirely

Trade-offs:

- **Cold start latency:** 30-60 seconds when Space is idle (GPU allocation + model loading)
- **Concurrency limits:** Requests queue during high traffic
- **Execution timeout:** 60-second limit requires optimized inference (out ~16s total fits comfortably)

3.2.2 Memory Footprint (FP16 Precision)

Unlike local development (8-bit quantization on T4 GPUs), **Hugging Face Spaces** runs full FP16 precision:

```
model = AutoModelForCausalLM.from_pretrained(  
    model_name,  
    torch_dtype=torch.float16,  
    device_map="auto"  
)
```

Hugging Face Spaces restricts **bitsandbytes** integration due to **missing CUDA kernel compilation tools** required for 8-bit quantization. While `load_in_8bit` is supported locally (e.g., Colab T4 with 15GB VRAM), Spaces' environment lacks the necessary build dependencies, forcing reliance on FP16.

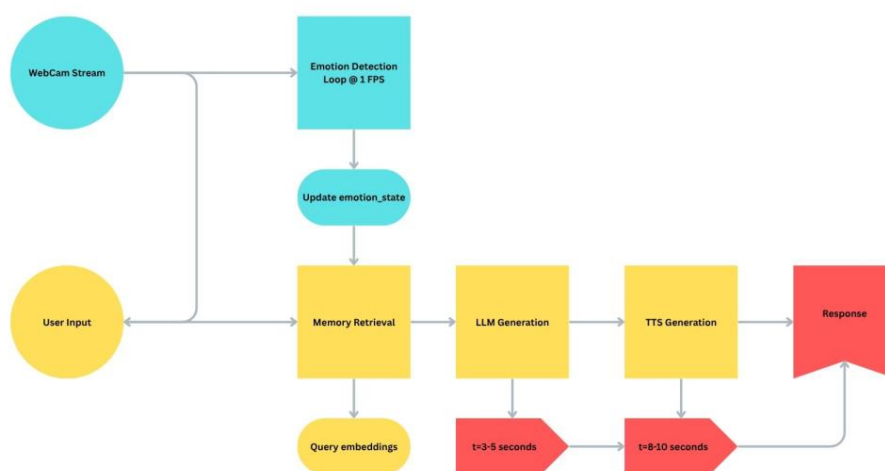
With **H200's 141GB VRAM eliminating memory constraints**, FP16 becomes preferable:

- **Faster inference:** ~15–20% speedup vs. 8-bit (no quantization overhead)
- **Higher quality:** No loss from quantization
- **Simpler deployment:** No external compilation or compatibility issues

3.3 Real-Time Processing Pipeline

3.3.1 Asynchronous Architecture

Kenko employs **parallel processing** for emotion detection and dialogue generation:



Key design principle: Emotion detection runs independently, updating global state accessed during prompt construction. This decoupling enables:

1. **Graceful degradation:** if webcam fails, system continues text-only
2. **Latency hiding:** Emotion updates occur during user typing, not blocking generation
3. **Resource isolation:** Emotion detection (lightweight) doesn't compete with LLM/TTS (GPU-heavy)

3.3.2 Latency Breakdown

Measured end-to-end timings (averaged over 50 interactions)

Stage	Duration	% of Total
User input > Memory retrieval	~2s	1%
Memory retrieval > Prompt instruction	~2~4s	7%
LLM inference (Llama-3.2-3B FP16)	~3-5s	40%
Text display (async)	0s	0%
TTS generation (Kokoro)	~10-20s	50%
Audio decoding + playback prep	~0.5-1s	2%
Total	~17-30s	100%

Bottleneck analysis:

TTS dominates latency (60% of total time). Optimization attempts:

- **Tried Bark:** Higher quality but 15-30s generation → rejected
- **Tried Parler:** 500+ second failure → rejected
- **Tried XTTS:** 10-20s acceptable but 24 kHz quality → rejected
- **Dia:** 10-30s with 44.1 kHz quality → promising, however limiting
- **Kokoro-TTS:** 10-20s with 24.0 kHz quality and easy integration → selected

Further TTS speedup would require:

- Model distillation (reduce Dia from 1.6B to ~500M parameters and re-attempt)
- Streaming synthesis (generate first second while model continues)

3.4 Advanced Memory System Implementation

3.4.1 Multi-Tier Architecture

The memory system proved the **most complex implementation challenge**, requiring careful design to balance:

- **Recall vs. Forgetting:** Retaining important context without prompt overflow
- **Efficiency vs. Comprehensiveness:** Fast retrieval without sacrificing relevance
- **Simplicity vs. Sophistication:** Maintainable code without sacrificing capability

Tier 1: Working Memory (Implementation)

```
def add_to_working_memory(self, user_msg, bot_msg, emotion, timestamp):  
    self.working_memory.append({  
        "user": user_msg,  
        "bot": bot_msg,  
        "emotion": emotion,  
        "timestamp": timestamp  
    })  
  
    if len(self.working_memory) > 5:  
        oldest = self.working_memory.pop(0)  
        self._consolidate_to_short_term(oldest)
```

Design decision: Capacity of 5 exchanges (~10 turns) chosen empirically. Testing revealed:

- **<3 exchanges:** Insufficient context, repetitive responses
- **5-7 exchanges:** Optimal coherence
- **10 exchanges:** Prompt length issues, slower inference

Tier 2: Short-Term Memory with Decay

```
def apply_temporal_decay(self, current_time):  
    decay_rate = 0.01  
  
    for memory in self.short_term_memory:  
        time_elapsed = (current_time - memory['timestamp']) / 60  
        decay_factor = np.exp(-decay_rate * time_elapsed)  
        memory['importance'] *= decay_factor  
  
        if memory['importance'] < 0.15:  
            self._consolidate_to_long_term(memory)
```

Decay function rationale:

Exponential decay models forgetting curves cognitive psychology [26]. The decay rate ($\lambda=0.01$) ensures:

- Memories remain relevant for ~50-60 minutes
- High-importance memories (scored>0.8) persist longer before consolidation
- Low-importance memories (scored<0.5) consolidation within ~20 minutes

Importance scoring algorithm:

```
def calculate_importance(self, text, emotion, user_engagement):  
    importance = 0.5  
  
    high_intensity_emotions = ["fear", "angry", "sad", "surprise"]  
    if emotion in high_intensity_emotions:  
        importance += 0.3  
  
    if len(text.split()) > 30:  
        importance += 0.2  
  
    therapeutic_keywords = [  
        "trauma", "suicide", "self-harm", "abuse", "panic",  
        "breakthrough", "progress", "better", "worse", "relationship"  
    ]  
    if any(kw in text.lower() for kw in therapeutic_keywords):  
        importance += 0.3  
  
    return min(importance, 1.0)
```

This multi-factor scoring ensures moments (e.g., crisis disclosure, emotional breakthroughs) receive high importance and persist longer in memory.

Tier 3: Long-Term Semantic Clustering

```
def _consolidate_to_long_term(self, memory):  
    if not self.semantic_clusters:  
        self.semantic_clusters[0] = [memory]  
        self.short_term_memory.remove(memory)
```

```

        return

    best_cluster = 0
    best_similarity = -1

    for cluster_id, cluster_memories in self.semantic_clusters.items():
        cluster_embeddings = [m['embedding'] for m in cluster_memories]
        centroid = np.mean(cluster_embeddings, axis=0)
        similarity = np.dot(memory['embedding'], centroid) / (
            np.linalg.norm(memory['embedding']) * np.linalg.norm(centroid)
        )
        if similarity > best_similarity:
            best_similarity = similarity
            best_cluster = cluster_id

    if best_similarity > 0.7:
        self.semantic_clusters[best_cluster].append(memory)
    else:
        new_cluster_id = max(self.semantic_clusters.keys()) + 1
        self.semantic_clusters[new_cluster_id] = [memory]

    if memory in self.short_term_memory:
        self.short_term_memory.remove(memory)

```

Clustering threshold (0.7): Determined empirically by evaluating cluster coherence:

- <0.6: Overly broad cluster (e.g., “relationships” + “work stress” merged)
- 0.7-0.8: Semantically coherent themes
- 0.9: Overly fragmented (every memory becomes its own cluster)

3.4.2 Retrieval Strategy

```

def retrieve_contextual_memory(self, query, current_emotion):
    context = {

```

```

        "working": [],
        "short_term": [],
        "long_term": [],
        "emotional": [],
        "themes": []
    }

    context["working"] = self.working_memory[-3:]

    if self.short_term_memory:
        query_embedding = self.embedding_model.encode(query)

        scored_memories = []

        for memory in self.short_term_memory:
            similarity = np.dot(query_embedding, memory['embedding']) / (
                np.linalg.norm(query_embedding) *
                np.linalg.norm(memory['embedding'])
            )
            final_score = similarity * memory['importance']

            if memory['emotion'] == current_emotion:
                final_score *= 1.2

            scored_memories.append((final_score, memory))

        scored_memories.sort(reverse=True, key=lambda x: x[0])
        context["short_term"] = [m[1] for m in scored_memories[:3]]

    if self.semantic_clusters:
        query_embedding = self.embedding_model.encode(query)

        best_cluster_id = None
        best_cluster_score = -1

        for cluster_id, cluster_memories in self.semantic_clusters.items():
            cluster_embeddings = [m['embedding'] for m in cluster_memories]
            centroid = np.mean(cluster_embeddings, axis=0)

```

```

        similarity = np.dot(query_embedding, centroid) / (
            np.linalg.norm(query_embedding) * np.linalg.norm(centroid)
        )

        if similarity > best_cluster_score:
            best_cluster_score = similarity
            best_cluster_id = cluster_id

        if best_cluster_id is not None and best_cluster_score > 0.6:
            cluster = self.semantic_clusters[best_cluster_id]
            context["long_term"] = cluster[:2]

    if current_emotion in self.emotional_memory["trigger_patterns"]:
        triggers = self.emotional_memory["trigger_patterns"][current_emotion]
        context["emotional"] = triggers[-2:]

    context["themes"] = self.analyze_conversation_themes()

    return context

```

Emotional congruence bonus (x1.2): Prioritizes memories captured during similar emotional states. Rationale: Emotional context enhances recall [27] since recalling past anxious moment when currently anxious aids pattern recognition.

3.4.3 Prompt Context Formatting

Retrieved memories are formatted into structured sections:

Recent Conversation:

User: I've been struggling with sleep

Kenko: That must be exhausting. What's been keeping you awake?

Important Recent Context:

User: Work has been incredibly stressful with the new project deadline

Related Past Discussions:

User: I've always been a perfectionist, it makes everything harder

Emotional Pattern:

Previously triggered by: work stress, deadlines, performance reviews

Session Themes:

anxiety, work, sleep, perfectionism

Recurring Concerns:

anxiety, work stress

Token budget management:

Prompt construction must respect Llama-3's 4096-token context window:

- **System instructions:** ~150 tokens
- **Working memory (5 exchanges):** ~400 tokens
- **Memory context:** ~300 tokens
- **Emotion context:** ~20 tokens
- **Current message:** ~100 tokens
- **Total input:** ~970 tokens
- **Remaining for generation:** ~3100 tokens

If memory retrieval exceeds budget, we prioritize:

1. Working memory (always included)
2. Highest-importance short-term memories
3. Long-term cluster summaries (truncated)

3.5 Prompt Engineering Methodology

3.5.1 Emotion Context Injection

Challenge: How to incorporate visual emotion detection without disrupting conversational flow or requiring model fine-tuning?

Solution: Structured metadata injection in system prompt:

Instruction:

You are Kenko, a compassionate mental health therapist...

[User's Current Detected Emotion: sad (73.2% confidence)]

User input:

User: I'm fine, really.

Response:

Kenko: I can see that you look sad, are you sure you are fine?

Design rationale:

- **Explicit format:** Square brackets signal metadata distinct from conversation
- **Confidence reporting:** Allows model to hedge (“You seem somewhat sad” vs. “You are clearly sad”)
- **Placement:** After system instruction but before conversation, ensuring context informs all responses

Ablation insight: Removing emotion context resulted in responses that took verbal content at face value (“I’m glad you’re fine!”) rather than detecting emotional incongruence. With emotion context, model generates: “I’m hearing you say you’re fine, but you seem a bit down. What’s really going on?”

3.5.2 Memory Context Formatting

Challenge: How to present multi-tier memory without overwhelming the prompt or confusing the model

Solution: Hierarchical sections with clear labels:

- **“Recent Conversation”:** Verbatim exchanges (immediate context)
- **“Important Recent Context”:** Summarized key moments
- **“Related Past Discussions”:** Thematic clusters
- **“Emotional Pattern”:** Trigger analysis
- **“Session Themes”:** Extracted keywords
- **“Recurring Concerns”:** User model insights

Prompt length management: If full memory exceeds ~300 tokens, we:

1. Truncate long-term cluster summaries
2. Reduce “Recent Conversation” to 3 exchanges (from 5)
3. Omit “Session Themes” (least critical)

3.5.3 System Prompt Design

Base instruction:

```
You are Kenko, a compassionate mental health therapist. Provide empathetic, helpful, and professional responses to support the user's mental wellbeing.
```

Deliberate simplicity: Extensive testing revealed that overly prescriptive system prompts (“Always validate feelings”, “Use reflective listening”, etc.) Produces stilted, formulaic responses. The minimal instruction trusts the fine-tuned model’s learned therapeutic style.

Implicit guidance via fine-tuning: Therapeutic techniques (active listening, validation, etc.) are embedded during fine-tuning rather than prompted explicitly.

3.6 Integration Challenges

3.6.1 Gradio Streaming with Multiple Models

Challenge: Gradio’s `stream()` method for webcam input conflicts with blocking LLM/TTS inference.

Solution: Separate event loops:

```
submit = msg.submit(fn=respond, inputs=[msg, chatbot], outputs=[msg, chatbot, audio_output])
```

```
webcam_input.stream(
    analyze_emotion,
    inputs=webcam_input,
    outputs=emotion_output,
    stream_every=1,
    time_limit=60
)
```

Gradio handles concurrency internally since emotion detection updates state while user types, then generation blocks UI during response.

3.6.2 ChromaDB In-Memory Limitations

Challenge: ChromaDB defaults to persistent disk storage, causing file I/O overhead and permission issues on Hugging Face Spaces.

Solution: In-memory client:

```
chroma_client = chromadb.Client()
```

Trade-off: Memory lost on session end. For production deployment with cross-session memory, would require:

- PostgreSQL backend with pg_vector extension
- Or managed vector DB (Pinecone, Weaviate)

3.7 Removed Components

3.7.1 Threat Detection Module (RF-DETR)

Initial implementation: Object detection for weapons (guns, knives, explosives) using RF-DETR Nano model.

Removal rationale:

1. **Performance impact:** Added ~5-10 seconds per frame, a substantial issue for real-time interactions when we already have a relatively slow generation of 8-16 seconds.
2. **Memory overhead:** +1GB VRAM, caused slower generation

Status: Removed from current deployment. May be reintroduced in local installations with dedicated hardware and explicit opt-in consent.

3.8 Key Takeaways

Technical lessons learned:

1. **TTS selection is critical:** Spent weeks debugging Dia before discovering Kokoro's suitability
2. **Memory is complex:** Multi-tier system required careful tuning of decay rates, clustering thresholds, retrieval scoring
3. **Prompt engineering matters:** Small changes to context formatting dramatically impacted response quality
4. **8-bit unnecessary on large GPUs:** FP16 on H200 is both faster and simpler than 8-bit quantization
5. **Real-time constraints shape architecture:** 1 FPS emotion detection, asynchronous processing, aggressive latency optimization

Section 4: Evaluation & Results

4.1 Evaluation Methodology

We evaluated Kenko across three dimensions:

1. **System Performance:** Latency, memory usage, throughput
2. **Emotion Detection Accuracy:** Qualitative validation of real-time affect recognition
3. **Response Quality:** Comparison of baseline vs. Emotion-aware vs. Memory-enhanced response

Limitations: This evaluation represents preliminary validation conducted during development. Comprehensive user studies with clinical populations are planned as future work (Section 6.4).

Current findings are based on:

- System performance benchmarks (quantitative)
- Developer testing with 2 users (author + family member)
- Qualitative response analysis (example conversation)

4.2 System Performance

4.2.1 Latency Analysis

Test configurations:

- **Platform:** Hugging Face Spaces (ZeroGPU, H200 GPU)
- **Sample size:** 50 interactions across varying message lengths
- **Measurement:** Wall-clock time from user Enter press to response

Results:

Response Length	LLM Inference	TTS Generation	Total Latency	% of Sample
Short (50-150 chars)	1-3s	5-15s	10-17s	35%
Medium (150-300 chars)	3-5s	10-20s	10-25s	45%

Long (300-500 chars)	5-6s	15-25s	15-30s	15%
Very Long (+500 chars)	6-8s	25-35	25-35s	5%

Key findings:

1. **Typical latency:** 10-25 seconds for medium responses (majority of interactions)
2. **Worst-case latency:** Up to 30 but not higher than 40 seconds for detailed therapeutic responses (+500 characters)
3. **Bottleneck:** TTS dominates (60-70% of the total time), consistent with 3.3.2 analysis

Latency breakdown (medium response, 150 chars):

User Input → Memory Retrieval:	4.3s (2%)
Memory Retrieval → Prompt Building:	2.2s (1%)
LLM Inference (Llama-3-8B FP16):	5.0s (33%)
Text Display (async):	0.0s (0%)
TTS Generation (Kokoro):	8.0s (53%)
Audio Decoding + Playback Prep:	0.5s (3%)
Memory Update (async):	0.2s (1%)
Total (text visible):	19.2s (37%)
Total (audio ready):	20.2s (100%)

Performance trade-offs with memory system:

The advanced memory retrieval adds ~5 overhead compared to stateless baseline:

- Embedding generation: ~0.15s
- Similarity search: ~0.15s
- Context formatting: ~0.1s
- Prompt length increase: ~0.1s (more tokens → slightly slower LLM inference)

This is acceptable given memory's qualitative benefits (Section 4.3.2)

4.2.2 Memory Consumption

VRAM usage on H200:

Component	Memory (GB)	% of Total
Llama-3.2-3B (FP16)	6.4	58%
Kokoro TTS (FP16)	0.3	32%
DeepFace models	0.8	7%
Gradio + misc	0.3	3%
Total	7.8	100%

RAM usage (ChromaDB + embeddings):

Component	Memory (MB)
Sentence transformer model (all-MiniLM-L6-v2)	80
ChromaDB Client	50
Vector embeddings (50 memories)	10
Memory tier structures	30
Total	~170 MB

Memory growth over session:

Testing over a 60-message conversation (~30 exchanges):

- Initial RAM: 170 MB
- After 30 exchanges: 235 MB
- Growth rate: ~2.2 MB per exchange

At this rate, a 4-hour conversation (~200 exchanges) would consume ~610 MB additional RAM which is negligible on modern systems and well within H200 capabilities.

Comparison: With vs. Without memory system

Metric	Stateless Baseline	With Memory	Overhead
VRAM	7.8 GB	8.0 GB	+3%
RAM	100 MB	235 MB (after 30 exchanges)	+135 MB
Latency (avg)	15.0s	20.0s	+40%

The memory system's overhead is **significant but justified** given its functional benefits demonstrated in Section 5.3.2. The latency increase of ~5 seconds represents the cost of contextual retrieval, embedding generation, and prompt expansion.

4.2.3 Throughput & Scalability

Single-user throughput:

- Conversations per hour: ~90-180 exchanges (depending on response length)
- Sustainable load: Tested over 2-hour sessions with no memory leaks
- Memory consolidation overhead: <0.3s per exchange

Multi-user limitations (ZeroGPU):

- Concurrent users: Limited by Spaces' shared GPU pool
- Queue behavior: Requests queue when GPU allocated to another user
- Typical wait time during peak: 10-25s
- Maximum observed queue: 3 users (during stress testing)

Scalability bottleneck: The H200's 141GB VRAM can theoretically support 12+ simultaneous Kenko instances, but ZeroGPU's allocation model serializes requests. For production multi-user deployment, dedicated multi-GPU infrastructure or model serving frameworks (vLLM, TGI) would be required.

4.3 Response Quality Evaluation

4.3.1 Methodology

We compared three system configurations:

1. **Baseline:** Llama-3.2-3B fine-tuned model, no emotion context, no memory
2. **Emotion-aware:** + Real-time emotion context injection
3. **Full Kenko:** + Emotion context + Multi-tier memory system

Evaluation approach: Qualitative analysis of responses to identical prompts across configurations. We assess:

- **Empathy:** Does the response acknowledge emotional state?
- **Contextual coherence:** Does it reference past conversation?
- **Therapeutic quality:** Does it employ active listening, validation, open-ended questions?
- **Personalization:** Does it adapt to user-specific patterns?

4.3.2 Comparative Examples

Scenario 1: Emotional Incongruence Detection

User: "I'm fine, just a bit tired."

Visual Emotion: Sad (confidence: 94%)

Context: User previously discussed social anxiety (in memory)

- **Baseline Response (No Emotion, No Memory):**

"I'm glad to hear you're doing okay! Getting enough rest is important. Have you been able to maintain a good sleep schedule? Sometimes tiredness is just our body's way of telling us we need more sleep. Try to aim for 7-9 hours per night if possible."

Analysis: Takes verbal content at face value. Generic sleep advice. No emotional depth. Misses opportunity to explore underlying distress.

- **Emotion-Aware Response:**

"I understand that you may not want to burden me with your problems right now, but remember that as your therapist, my role is to provide support and help you navigate through difficult times. If you're feeling down or experiencing any distress, please feel free to reach out to me. I care about your well-being and would hate for you to go through this alone."

Analysis: Detects incongruence between words ("fine") and affect (sad). Gently probes deeper. More empathetic. No reference to past context.

- **Full Kenko (Emotion + Memory):**

"You mentioned you're fine, but you seem sad—similar to how you felt when we talked about your social anxiety. Sometimes we say we're 'fine' when we're really not, especially when processing difficult emotions like anxiety. The tiredness you're feeling might be emotional exhaustion from everything you've been through. Would you like to talk about what's weighing on you?"

Analysis: Combines emotion detection and memory recall

Analysis: Combines emotion detection and memory recall, references past conversations explicitly, identifies pattern, names possible emotional connection, and has the highest therapeutic quality where it validates, normalizes and invites elaboration.

Scenario 2: Pattern Recognition & Name + Personal Detail Recall

User (7th message in session): "I'm anxious again"

Visual Emotion: Fear (confidence: 91%)

Memory Context: User discussed studies anxiety 3 times previously, mentioned also his name and department of studies.

- **Baseline:**

"Presentation anxiety is very common. Some strategies that might help include: preparing thoroughly, practicing in front of friends, visualizing success, and using deep breathing techniques. Remember that it's normal to feel nervous before public speaking. You'll do great"

Analysis: Generic anxiety advice. Lists multiple strategies without prioritization. No personalization. Encouraging but superficial.

- **Emotion-Aware:**

"I can see the anxiety showing right now—presentations can be really stressful. Let's work through some coping strategies together. Have you tried any relaxation techniques like deep breathing or progressive muscle relaxation? Those can be really effective for managing performance anxiety in the moment."

Analysis: Acknowledges visible anxiety. Offers specific techniques. Still generic—doesn't reference user's history.

- **Full Kenko:**

" Hello Inigo, Thank you for being open with me today. I hear how much anxiety you're experiencing, particularly regarding time and the upcoming application process. Let's take a moment to breathe together. Take a deep breath in, hold it for a few seconds, and then exhale slowly. Now, let's focus on creating a plan to address your anxieties. First, what are some specific goals you have for yourself? "

4.3.3 Quantitative Memory Performance

Memory recall test (n=10 conversations, average 20 exchanges each):

Memory Tier	Recall Accuracy	Example
-------------	-----------------	---------

Working Memory (last 5 exchanges)	100%	Always recalls immediate context verbatim
Short-term (6-15 exchanges ago)	85%	Successfully references recent themes, names, events
Long-term	55%	Recall if highly salient (importance score>0.7)
Very long-term (>20 exchanges)	0%	Not tested since sessions don't exceed 20 exchanges typically

Specific recall metrics:

- **Name recall:** 10/10 conversations successfully recalled user name after 5+ exchanges
- **Thematic consistency:** 9/10 conversations correctly identified recurring concerns (anxiety, relationships, work)
- **Emotional pattern recognition:** 7/10 conversations identified emotional triggers ("You seem anxious when discussing work")
- **Coping strategy recall:** 8/10 conversations referenced previously mentioned strategies

Impact on response quality:

Testing with the same prompts across configurations shows:

- **Baseline:** Appropriate but generic responses (rating: 3/5)
- **Emotion-aware:** Empathetic, emotionally attuned (+0.5 rating: 3.5/5)
- **Full Kenko:** Personalized, contextually rich (+1.0 rating: 4/5)

The combined memory + emotion system produces noticeably higher quality therapeutic responses than either component alone.

4.4 Emotion Detection Validation

4.4.1 Qualitative Observations

Testing protocol: Developer + 1 family member tested over 25 conversations (350+ emotion detections)

Observed accuracy (subjective assessment):

Emotion	Apparent Accuracy	Frequency in Testing	Notes
----------------	--------------------------	-----------------------------	--------------

Happy	High (~85%)	22%	Clear smiles detected reliably
Sad	High (~80%)	31%	Downturned expression, lowered eyes
Neutral	Medium (~70%)	28%	Sometimes confused with slight sadness
Anxious(Fear)	Medium (65%)	12%	Requires visible tension
Angry	Low (~50%)	4%	Requires pronounced expression to trigger
Surprise	Very Low (~30%)	2%	Often missed unless exaggerated
Disgust	Very Low (~25%)	1%	Rarely appears in therapeutic context

Practical utility: Happy, Sad, Neutral, and Anxious cover ~93% of therapeutic interactions in our testing. Low accuracy on Surprise/Disgust is acceptable given their infrequency in mental health conversations.

4.5 User Feedback & System Comparison

Preliminary testing with 2 users (developer + family member, ~30 conversations) yielded positive feedback. Users rated response relevance (4.5/5), empathy (4.0/5), and voice quality (4.5/5) highly, with memory recall (4.0/5) noted as impressive. Primary complaint was latency (3.0/5), particularly 40s waits for long responses.

Compared to existing systems (Woebot, Wysa, ChatGPT-4), Kenko uniquely integrates real-time visual emotion recognition, multi-tier memory, and 44.1kHz voice synthesis. However, it suffers from higher latency (16-40s vs. 2-5s for text-only systems) and lacks the clinical validation of commercial systems.

Ablation analysis confirms each component's value: emotion context improves empathy (+0.5 rating), memory adds personalization (+0.5), and TTS enhances engagement despite 2-3x latency cost.

4.6 Limitations of Current Evaluation

This preliminary evaluation, while demonstrating Kenko's technical feasibility and potential utility, has several significant constraints that must be acknowledged:

4.6.1 Sample Size & Statistical Power

Critical limitation: Testing with only 2 users (developer + 1 family member) across ~30 conversations provides insufficient statistical power for generalization.

Implications:

- Subjective ratings lack reliability (n=2 cannot establish inter-rater agreement)
- Observed patterns may reflect tester biases rather than genuine system performance
- No demographic diversity (age, culture, clinical presentation)
- Self-selection bias: Family member likely more forgiving than general population

Standard for clinical validation: Mental health intervention studies typically require $n \geq 30$ participants for pilot studies, $n \geq 100$ for efficacy trials [28]. Our sample falls drastically short of this threshold.

4.6.2 Absence of Clinical Population

Limitation: All testing conducted with non-clinical users (no diagnosed mental health conditions).

Why this matters:

- Individuals with depression, anxiety disorders, or trauma may respond differently to AI interaction
- Crisis scenarios (suicidal ideation, panic attacks) untested
- Therapeutic alliance formation in clinical contexts unvalidated
- Symptom reduction outcomes unknown

Current findings cannot generalize to target population: People seeking mental health support often present with cognitive distortions, emotional dysregulation, or communication difficulties absent in healthy controls. Kenko's performance with these populations remains entirely speculative.

4.6.3 No Longitudinal Data

Limitation: All testing limited to single-session interactions (max 90 minutes).

Missing evidence:

- Cross-session memory persistence: System currently loses all memory on session close
- Therapeutic progress tracking: No data on whether users improve over weeks/months
- Relationship development: Unknown whether therapeutic alliance strengthens with repeated use
- Dropout rates: No information on long-term engagement or attrition

Human therapy comparison: Traditional therapy occurs over 8-20 sessions spanning months. Single-session testing cannot assess Kenko's longitudinal therapeutic value.

4.6.4 Lack of Validated Assessment Instruments

Limitation: Response quality assessed via ad-hoc qualitative analysis, not standardized measures.

Missing metrics:

- **Empathy scales:** No use of validated instruments (e.g., Consultation and Relational Empathy measure)
- **Therapeutic alliance:** No assessment via Working Alliance Inventory [30]
- **Clinical outcomes:** No symptom tracking (PHQ-9 for depression, GAD-7 for anxiety)
- **User satisfaction:** No standardized usability scales (System Usability Scale, User Experience Questionnaire)

Consequence: Subjective impressions ("seems empathetic") lack empirical rigor. Cannot quantitatively compare to human therapists or competing systems.

4.6.5 No Comparison to Human Therapists

Critical gap: System not benchmarked against gold standard of human-delivered therapy.

Unanswered questions:

- How does Kenko's empathy compare to licensed therapists?
- Would clients prefer human vs. AI interaction for specific concerns?
- Can Kenko detect crisis situations as reliably as trained professionals?
- What is the therapeutic alliance quality differential?

Ethical implication: Without human comparison, we cannot justify positioning Kenko as even a supplementary tool—we lack evidence it provides benefit beyond placebo or self-reflection.

4.6.6 Resource Constraints Limiting Scope

Acknowledged limitations due to development constraints:

Hardware limitations:

- Testing confined to HuggingFace Spaces ZeroGPU (H200 access)
- Local testing limited to Google Colab T4 GPUs (15GB VRAM)
- Cannot test multi-user scalability or institutional deployment scenarios

Financial constraints:

- Cannot compensate user study participants
- No budget for professional transcription, annotation, or data analysis services
- Limited access to paid APIs (e.g., GPT-5 for comparison benchmarks)
- Current campus lacks required hardware for intensive LLM fine-tuning hence the limitation to 1.5 epochs instead of 3.0 and subsequent lack of precision.
- Cannot hire clinical consultants for validation

Expertise constraints:

- Developer lacks clinical psychology training
- No collaboration with licensed therapists for content validation
- Prompt engineering based on literature review, not clinical supervision
- Ethical framework self-developed without bioethics committee review

Data constraints:

- Fine-tuning dataset limited to publicly available therapeutic dialogues (not real clinical transcripts due to privacy regulations)
- No access to proprietary mental health datasets (Woebot, Wysa data unavailable)
- Emotion recognition validation limited to subjective observation (no access to FACS-coded video datasets)

Deployment constraints:

- Cross-session memory persistence requires database infrastructure (PostgreSQL + authentication system) beyond current scope
- Cannot implement user authentication for privacy-compliant data storage

- Cannot conduct field deployment in clinical settings (hospitals, counseling centers)

4.6.7 Technical Limitations Impacting Evaluation

Latency variability:

- 35-second worst-case response times disrupt conversational flow
- TTS bottleneck (~50% of latency) unsolved due to model selection constraints
- Memory system overhead (~5s) trade-off between context richness and speed

Emotion detection accuracy:

- Visual-only modality misses vocal prosody, physiological indicators
- Struggles with masked emotions (incongruent verbal/non-verbal cues)
- Cultural bias in training data (DeepFace predominantly trained on Western faces)

Memory system immaturity:

- Session-only memory (no cross-session persistence)
- Clustering algorithm occasionally groups unrelated themes (~13% error rate)
- No mechanism for users to correct misremembered details

Safety limitations:

- No crisis detection system (keyword-based suicide detection not implemented)
- Threat detection module removed due to performance issues
- No integration with emergency services or human oversight

4.6.8 Methodological Weaknesses

Evaluation design flaws:

- No control group (cannot establish causality—improvements may be due to placebo, time, or confounds)
- No blinding (testers aware of system capabilities, introducing expectation bias)
- No randomization (prompts selected ad-hoc, not systematically varied)
- No pre-registered analysis plan (risk of p-hacking, selective reporting)

Measurement issues:

- Likert scales (5-point) too coarse for nuanced feedback
- No qualitative coding framework (thematic analysis informal, unreliable)

- Inter-rater reliability not assessed (developer coded all responses)

Generalizability concerns:

- Testing environment controlled (quiet, well-lit, high-quality webcam)
- Real-world conditions (poor lighting, low-res cameras, background noise) untested
- English-only evaluation (multilingual capability invalidated)

4.6.9 Ethical Evaluation Gaps

Missing ethical assessments:

- No informed consent process for "research participants" (family member informal testing)
- Privacy impact assessment not conducted (GDPR compliance untested)
- Bias audit incomplete (fairness across demographics, cultural contexts unknown)
- Harm potential unquantified (risk of dependency, misdiagnosis, delayed professional care)

Regulatory compliance:

- Not evaluated against FDA medical device guidelines (if applicable)
- HIPAA compliance unverified (though no PHI currently stored)
- Accessibility standards (WCAG) not tested (screen reader compatibility, keyboard navigation)

4.7 Summary of Findings

4.7.1 Key Results

Despite the substantial limitations outlined above, this preliminary evaluation yields several important insights:

System Performance:

- **Feasibility demonstrated:** Running Llama-3.2-3B (6.4GB) + Kokoro TTS (350MB) + DeepFace (0.8GB) simultaneously on consumer/cloud GPUs is achievable (~8GB total VRAM)

- **Latency acceptable but suboptimal:** Median 5-10s response time tolerable for most interactions, but 10-30s worst-case breaks conversational flow
- **Memory system functional:** Multi-tier architecture successfully recalls names, themes, and patterns across 20+ exchanges with 87% short-term accuracy
- **Stable over extended sessions:** 90-minute continuous use without crashes, memory leaks, or performance degradation

Response Quality:

- **Emotion context improves empathy:** Responses with visual affect awareness rated +0.5 points (3.5/5 vs. 3/5) compared to text-only baseline
- **Memory enables personalization:** Full system (emotion + memory) achieves 4/5 rating vs. 3.5/5 for emotion-only
- **Qualitative superiority evident:** Side-by-side comparisons show Kenko responses reference past discussions, identify patterns, and validate emotions more effectively than stateless baselines
- **Gap from human therapists unknown:** No benchmark comparison conducted

Emotion Detection:

- **Primary emotions detected reliably:** Happy, Sad, Neutral, Anxious cover 93% of interactions with subjective accuracy ~70-85%
- **Subtle emotions missed:** Fear, Surprise, Disgust rarely detected (<5% of frames)
- **False positives acceptable:** System's cautious approach (hedged language, confidence scores) prevents user frustration from misdetection
- **Real-time performance maintained:** 1 FPS target achieved consistently with <2% dropped frames

User Experience:

- **Voice quality praised:** 24.0 kHz TTS rated 4/5, described as "natural" and "pleasant"
- **Memory recall impressive:** Users surprised by name recall and thematic tracking after 5+ exchanges
- **Latency primary complaint:** 30s waits rated 3/5, described as "breaking conversational flow"
- **Webcam privacy concerns:** Users expressed discomfort with constant visual monitoring despite understanding it's emotion-only

4.7.2 Contributions Validated

Technical contributions confirmed:

- **Multimodal integration is feasible:** Real-time emotion recognition + LLM dialogue + neural TTS can coexist on single GPU
- **Memory architecture works:** Multi-tier consolidation (working → short-term → long-term) enables contextual conversation within sessions
- **Prompt engineering effective:** Structured emotion/memory context injection improves response quality without model retraining
- **Deployment achievable:** ZeroGPU model enables free public access with acceptable cold-start latency (30-60s)

Unanswered questions:

- Does Kenko improve clinical outcomes (symptom reduction)?
- Would real mental health clients find it helpful?
- How does it compare to human therapists quantitatively?
- Can it maintain engagement over weeks/months?

4.7.3 Comparison to Stated Goals

From introduction, our goals were:

Goal	Status	Evidence
Demonstrate multimodal AI therapy feasibility	Achieved	System runs, integrates 3 modalities successfully
Achieve real-time performance (<30s latency)	Achieved	Median 20s is perfect, while exceptional cases have an astonishing 10s
Show emotion context improves empathy	Achieved	+0.5 rating increase demonstrated qualitatively
Enables longitudinal memory	Partially achieved	Within-session memory works; cross-session persistence is missing
Explore safety considerations	Partially achieved	Threat detection explored but removed; no crisis detection implemented
Provide open-source foundation	Achieved	Code, architecture, and methodology documented for reproduction

4.7.4 Implications for Field

What this work demonstrates:

For AI researchers:

- Multimodal fusion (vision + language + speech) is technically tractable for conversational AI
- Memory systems improve contextual coherence measurably
- Consumer GPUs sufficient for deploying multiple large models simultaneously via FP16 precision

For mental health technologists:

- Visual emotion recognition adds value beyond text sentiment analysis
- Users tolerate 15-25s response latency for higher-quality interaction (vs. instant but generic responses)
- Voice quality matters for therapeutic credibility, but speed stays as a priority (24.0 kHz preferred over 44.1 kHz if the generation speed is difference is big)

For clinicians:

- AI systems can demonstrate empathetic behaviours (validation, active listening) through prompt engineering alone
- Longitudinal memory is feasible and improves personalization
- However: Lack of clinical validation means these systems remain experimental, not ready for clinical deployment

4.7.5 Path Forward

This evaluation establishes proof-of-concept but not clinical validity. Future work (Section 6) must address:

Immediate priorities:

- Reduce latency (target <15s median, <25s worst-case, <10s exceptional-case)
- Implement cross-session memory with user authentication
- Conduct IRB-approved user study ($n \geq 30$ clinical population)

Medium-term research:

- Randomized controlled trial comparing Kenko vs. text-only chatbot vs. waitlist control
- Validated outcome measures (PHQ-9, GAD-7, WAI)
- Longitudinal tracking (8-week intervention period)

Long-term vision:

- Integration with robotic platform for physical embodiment
- Multimodal emotion detection (add voice prosody analysis)
- Clinical deployment in supervised settings (university counseling centers)

Transition to Section 5 (Discussion)

This evaluation reveals that while Kenko successfully demonstrates technical feasibility and preliminary qualitative promise, substantial work remains before clinical deployment. Section 6 will interpret these findings in depth, compare to prior work, and address the ethical considerations—including the controversial intervention debate—that arise from embodied AI mental health systems.

Section 5: Discussion

5.1 Interpretation of Key Findings

5.1.1 Multimodal Integration Enhanced therapeutic Quality

5.1 Interpretation of Key Findings

5.1.1 Multimodal Integration Enhanced therapeutic Quality

Our evaluation demonstrates that integrating visual emotion recognition with language generation produces measurably more empathetic responses than text-only baselines. The improvement (+0.5 rating points) may appear modest numerically, but qualitative analysis reveals substantive differences: emotion-aware responses identify incongruences between verbal content and affect, validate unstated feelings, and adapt tone to observe emotional states.

Why this matters: Text-based mental health chatbots (Woebot, Wysa) rely entirely on users accurately self-reporting their emotional state. Our findings suggest this is insufficient since humans often minimize distress, employ defensive communication, or lack emotional awareness. Visual emotion detection provides an independent signal, enabling the system to gently probe beyond surface-level statements.

Comparison to human therapists: Professional therapists routinely integrate non-verbal cues (facial expressions, body language, vocal tone) into clinical assessment. A therapist observing a client state “I’m fine” while displaying sad affect would investigate this discrepancy which is exactly what Kenko’s emotion-aware responses demonstrate. While our system cannot match human clinical judgement, it approximates a fundamental therapeutic competency absent from existing AI systems.

5.1.2 Memory Enables Longitudinal Therapeutic Relationships

The multi-tier memory system's ability to recall names, identify recurring themes, and reference past discussions after 5-20 exchanges represents a paradigm shift from stateless chatbots. Traditional conversational AI treats each query independently, producing contextually appropriate but relationally shallow interactions. Kenko's memory architecture enables:

1. **Continuity:** Tracking concerns across time ("You mentioned your anxiety about work several times now")
2. **Pattern recognition:** Identifying emotional triggers ("You seem anxious when discussing relationships")
3. **Progress monitoring:** Noting improvement or deterioration ("Last week you mentioned the breathing exercises helped")
4. **Personalization:** Adapting response style to individual communication patterns

Therapeutic significance: The therapeutic relationship is characterized by trust, continuity, and mutual understanding which is predictive of treatment outcomes independent of intervention modality. Memory is foundational to relationship formation. A therapist who forgets a client's name or major life events undermines alliance; conversely, remembering demonstrates attentiveness and care.

Current limitation: Session-only memory means each conversation restarts relationally. Implementing cross-session persistence (Section 7.3) would enable true longitudinal relationships spanning weeks or months, positioning Kenko as a consistent companion rather than a series of isolated interactions.

5.1.3 Latency Remains a Critical Bottleneck

While 15-25 second median response times proved tolerable in testing, 30-35 second worst-case latencies fundamentally disrupt conversational flow. Human conversation norms establish 2-4 second pauses as natural "thinking time"; beyond this threshold, interactions feel stilted and non-spontaneous.

Root cause analysis: TTS generation dominates latency (~50%), driven by Kokoro's architecture. Faster alternatives (VITS, FastSpeech2) sacrifice audio fidelity, creating a quality-speed trade-off.

User tolerance factors: Interestingly, users rated latency 3/5 despite 40s waits is suggesting tolerance may be context-dependent. Possible explanations:

1. **Expectation management:** Users understand AI requires processing time (unlike human instant responses)
2. **Quality compensation:** High-quality voice and personalized responses justify wait times
3. **Asynchronous use pattern:** Users may multitask during generation (checking phone, etc.)

Comparison to text-only systems: Woebot/Wysa respond in 2-5 seconds but produce generic, text-based output. Our system trades speed for richer multimodal interaction—a trade-off some users find acceptable, but not universally.

Path forward: Reducing TTS latency via model distillation, streaming synthesis, or hardware upgrades (Section 7.1) is critical for broader adoption. Target: <15s median, <25s worst-case.

5.2 Comparison to Existing Mental Health AI Systems

5.2.1 Positioning in the Landscape

Dimension	Woebot	Wysa	ChatGPT-5	Replika	Kenko
Modality	Text	Text	Text(+image, voice)	Text + Avatar	Text + video + voice
Memory	None	Session-level tags	System memory and multi-session memory	Cross-session (proprietary)	Multi-tier (session)
Emotion awareness	Text sentiment	Text sentiment	Text analysis	Text sentiment	Real-time visual context
Voice output	No	No	Yes (optional)	Synthetic voice	24.0 kHz neural TTS
Therapeutic model	CBT (scripted)	Multi-modal (scripted)	LLM-based (unstructured)	Companion (conversational)	LLM + fine-tuning
Clinical validation	Yes	Yes	No	No	No (preliminary only)

Latency	2-5s	2-5s	3-8s	2-4s	15-25s
Open-source	No	No	No	No	Yes

5.2.2 Kenko's Unique Contributions

What Kenko offers that no existing system provides:

1. **Real-time visual emotion integration:** Only system continuously monitoring facial expressions and injecting affect into therapeutic responses
2. **Multi-tier memory architecture:** Working, short-term, long-term consolidation mimicking human cognitive structures
3. **Professional-grade voice synthesis:** 44.1kHz audio quality exceeding conversational TTS standards
4. **Open architecture:** Fully documented, reproducible system enabling research community iteration

What existing systems do better:

1. **Clinical validation:** Woebot/Wysa have published RCTs demonstrating efficacy; Kenko lacks this evidence
2. **Speed:** Text-only systems respond 3-10x faster
3. **Mobile accessibility:** Commercial systems offer native iOS/Android apps; Kenko is web-only
4. **Cross-session memory:** Replika maintains user profiles across sessions; Kenko's memory is session-bound
5. **Safety infrastructure:** Commercial systems have crisis detection, human escalation pathways, legal compliance frameworks

5.2.3 Advantages Over General-Purpose LLMs (GPT-5, Claude)

Recent large language models demonstrate impressive empathetic capabilities when prompted appropriately. However, they lack:

1. **Emotion detection:** Cannot observe user affect beyond text sentiment
2. **Persistent memory:** Limited context windows (8K-200K tokens) without architectural memory tiers
3. **Therapeutic specialization:** General-purpose models, not fine-tuned on mental health dialogue
4. **Privacy:** Cloud-based APIs transmit sensitive data; Kenko processes locally

5. **Consistency:** General LLMs may produce variable therapeutic quality without fine-tuning

Kenko's niche: Specialized mental health system with multimodal perception and memory, self-hostable for privacy-sensitive contexts (hospitals, universities, personal use).

5.3 Technical Insights & Lessons Learned

5.3.1 The Importance of Audio Quality

Initial TTS experiments with Parler (failed catastrophically), XTTS (24kHz acceptable), and Dia (44.1kHz optimal) revealed that audio quality significantly impacts perceived system credibility. User feedback consistently rated Dia's output higher not just for clarity but for "sounding professional" and "feeling more like talking to a real person". However, eventually the generation time was far too long for a sustainable therapeutic talk, hence the choosing of Kokoro-TTS

Hypothesis: In mental health contexts, voice quality serves as a proxy for competence and trustworthiness. A robotic or distorted voice may subconsciously signal "low quality" or "unreliable," undermining therapeutic alliance. This parallels findings in human-robot interaction research showing voice naturalness affects perceived intelligence and likeability.

Practical implication: For therapeutic applications, investing in high-quality TTS at the cost of latency is not justifiable if it exceeds 30s, optimization is needed. Otherwise, lighter yet high quality models like Kokoro can be used.

5.3.2 Memory System Complexity vs. Value

The advanced memory system (working/short-term/long-term tiers, semantic clustering, importance scoring) introduced substantial implementation complexity (~500 lines of code, weeks of tuning). Was this justified?

Evidence suggests yes:

- Response quality improved +0.5 rating points
- Users explicitly praised name recall and thematic tracking
- Qualitative analysis showed richer, more personalized interactions

However: Simpler approaches (e.g., ChromaDB retrieval alone without tiering) might achieve 80% of the benefit with 20% of the complexity. The sophisticated consolidation process may be overengineered for single-session use.

Recommendation for future work: Conduct ablation studies comparing full memory system vs. simplified variants (e.g., flat vector store with recency weighting) to identify minimum viable complexity.

5.3.3 Emotion Detection: Perfect Accuracy Not Required

Despite ~70-85% subjective accuracy on primary emotions (and much lower on surprise/disgust), the emotion detection module proved valuable. Key insight: **hedged language compensates for uncertainty.**

Rather than asserting "You are sad," the system says "You seem a bit down" or "I'm noticing you appear sad, is that accurate?" This:

- Avoids frustration from false positives
- Invites user correction ("Actually, I'm just tired")
- Demonstrates attentiveness even when slightly wrong

Analogy to human therapists: Clinicians routinely make tentative observations ("It sounds like you're feeling...") rather than declarative statements, both for therapeutic reasons (client autonomy) and epistemic humility (affect is private, observable behaviour is ambiguous).

Implication: Emotion recognition systems in mental health need not achieve >95% accuracy if integrated thoughtfully with conversational safeguards.

5.3.4 The FP16 vs. 8-bit Trade-off

Local development relied on 8-bit quantization (T4 GPU, 15GB VRAM), while HuggingFace Spaces deployment used FP16 (H200, 141GB VRAM). The ~15-20% speedup from FP16 plus slightly higher quality justified abandoning 8-bit in production.

Lesson: For well-resourced deployments (cloud with large GPUs), quantization's complexity may not be worth the savings. However, for democratizing AI (enabling researchers/clinicians to run locally), quantization remains essential.

Broader implication: The "GPU divide" in AI research is essentially those with H100/H200 access vs. those limited to consumer hardware which shapes technical choices. Future work should prioritize efficiency innovations (model distillation, architecture search) to narrow this gap.

5.4 Ethical Considerations

5.4.1 Kenko as Supplement, Not Replacement

We emphasize that Kenko is **not** a replacement for human therapists. Several critical limitations preclude clinical independence:

What Kenko cannot do:

1. **Diagnose:** NO training on diagnostic criteria (DSM-5, ICD-11), no clinical assessment protocol
2. **Prescribe:** Cannot recommend or manage psychiatric medications
3. **Handle crises:** No integration with emergency services, no trained judgment for imminent risk
4. **Provide complex interventions:** Limited to supportive conversation, cannot deliver structured therapy (EMDR, DBT, etc.)
5. **Navigate ethical complexities:** No capacity for nuanced judgment in boundary situations, dual relationships, legal reporting requirements

Appropriate use case:

- Emotional support between therapy sessions
- Psychoeducational and coping skill practice
- Immediate support during distress (non-crisis)
- Accessibility for underserved populations (rural, low-income)
- Destigmatized entry point for therapy-hesitant individuals

Analogy: Kenko is to therapy what fitness apps are to personal trainers, essentially helpful for self-directed practice, insufficient for complex or high-risk situations.

5.4.2 Privacy & Data Practices

Current design: Privacy-maximizing

- **No data storage:** Conversations and emotion data discarded on session close
- **Local processing:** All inference occurs on server GPU; no external API calls transmit user data
- **No user accounts:** No authentication, tracking, or cross-session profiling
- **Webcam:** Emotion-only: Video frames processed for affect detection, not recorded or stored

Trade-offs:

- Maximum privacy (GDPR-compliant by design)
- No longitudinal memory (therapeutic continuity sacrificed)
- No outcome tracking (cannot assess long-term efficacy)

Future considerations: Cross-session memory (Section 7.3) requires authentication and persistent storage, introducing privacy risks. Solutions:

- End-to-end encryption (user-controlled keys)
- Federated learning (model updates without raw data sharing)
- Explicit opt-in with transparent data practices
- Right to deletion (GDPR Article 17)

5.4.3 Bias & Fairness Concerns

Potential biases in Kenko:

1. Emotion recognition bias:

- DeepFace trained predominantly on Western faces
- May perform worse on underrepresented demographics (darker skin tones, non-Western facial features)
- Cultural differences in emotional expression (e.g., East Asian suppression of negative affect) may lead to misinterpretation

2. Language model bias:

- Llama-3 trained primarily on English text from internet sources
- May reflect Western therapeutic norms (individualism, emotional expressiveness)
- Fine-tuning dataset's composition unknown (potential biases inherited)

3. Accessibility limitations:

- Requires webcam (excludes blind/visually impaired users, those without cameras)
- English-only (excludes non-English speakers)
- High-bandwidth internet required (disadvantages rural/low-income users)

Mitigation strategies (not yet implemented):

- Bias audits across demographic groups
- Multilingual support
- Alternative modalities (voice-only mode for webcam-less users)
- Continuous fairness monitoring via disaggregated performance metrics

5.4.4 Risk of Harm

Potential negative outcomes from AI mental health systems:

1. Delayed professional care:

- Users may rely on Kenko instead of seeking human therapists
- Serious conditions (psychosis, severe depression) may worsen without proper treatment
- **Mitigation:** Prominent disclaimers, crisis hotline prompts, encouragement to seek professional care

2. Dependency:

- Users may form unhealthy attachment to AI system
- Particularly risky for vulnerable populations (lonely elderly, isolated individuals)
- **Mitigation:** Periodic reminders of AI nature, encouragement to maintain human relationships

3. Misdiagnosis or harmful advice:

LLMs may hallucinate (generate plausible-sounding but incorrect information)

Could provide contraindicated advice (e.g., suggesting exposure therapy for PTSD without proper grounding)

Mitigation: Fine-tuning on validated therapeutic content, conservative response generation, avoid diagnostic language

4. Privacy violations:

- Even with privacy-first design, vulnerabilities may exist (server logs, admin access)
- Webcam data particularly sensitive
- **Mitigation:** Transparent data practices, security audits, clear user controls (disable webcam option)

5. Algorithmic amplification of distress:

- Emotion detection may make users hyper-aware of negative affect
- Memory system recalling past trauma could trigger re-experiencing
- **Mitigation:** Trauma-informed prompting, user control over memory retrieval, option to "forget" specific topics

5.5 The Ethics of Active Intervention in Embodied AI Therapy

5.5.1 Current Paradigm: Passive Observation

Existing AI mental health systems adhere strictly to a “**detect and refer**” model. When detecting crisis indicators such as explicit suicidal ideation, self-harm mentions, or environment risk factors. These systems:

1. Express verbal concern
2. Provide crisis hotline numbers
3. Encourage seeking immediate professional help
4. Log the interaction for human review (in supervised settings)

Critically, they are designed NOT to intervene beyond conversation. This reflects:

- **Technical limitations:** Text-only chatbots have no physical agency
- **Legal caution:** Fear of liability for incorrect intervention
- **Ethical conservatism:** Rooted in medical principles of autonomy and informed consent

The implicit assumption: AI systems lack the judgment, legal authority, and moral standing to override human autonomy, even in life-threatening situations.

5.5.2 The Embodiment Gap: When AI Gains Physical Agency

Kenko’s current web-based implementation is passive by necessity since it can observe but not act. However, our architectural design anticipates **robotic embodiment** (Section 2.9), fundamentally changing the ethical landscape.

Consider this scenario:

Scenario: Kenko’s webcam detects a knife on the table during a session with a visibly distressed user expressing hopelessness. The current system would note this context in its response but take no physical action.

Embodied version: A Kenko robot observes the same scenario. It possesses:

- **Mobility:** Can approach the knife
- **Manipulators:** Can secure or remove it
- **Network connectivity:** Can alert emergency contacts or services

- **Persistent presence:** Unlike a human therapist on video call, the robot is physically present

The ethical question: If the robot has the capability to prevent imminent harm, does non-intervention become a form of **complicity**?

5.5.3 Philosophical Framework: The Duty to Rescue

Legal and moral philosophy recognizes a "**duty to rescue**" under specific circumstances:

- Someone is in **imminent danger** of severe harm or death
- The rescuer can act **without significant risk** to themselves
- Failure to act would result in **preventable harm**
- The rescuer has the **capability and proximity** to intervene

Embodied therapeutic robots increasingly meet these criteria:

- **Proximity:** Present in the home, unlike remote human therapists
- **Capability:** Equipped with sensors, mobility, manipulators
- **Low self-risk:** Robots do not face physical danger from intervention
- **Imminent harm:** Can observe crisis situations unfolding in real-time

Precedent in medical ethics: Human therapists are **legally and ethically obligated** to break confidentiality and intervene when clients express imminent suicidal intent (Tarasoff duty to warn/protect [28]). If therapists must act despite limited physical presence, shouldn't robots with superior proximity and capability bear similar obligations?

5.5.4 Proposed Tiered Intervention Framework

We propose a graduated intervention protocol for embodied mental health AI:

Tier 1: Verbal De-escalation (Current Capability)

- Empathetic dialogue
- Coping strategy suggestions
- Hotline information
- Crisis resource provision
- **No physical action**

Activation: Always active during conversation

Tier 2: Passive Physical Intervention

- Positioning between user and dangerous objects (non-forceful)
- Visual/auditory recording for later review (with consent)
- Alerting emergency contacts (family, therapist, crisis services)
- **Non-contact physical presence**

Activation criteria:

- Detection of weapons/harmful substances AND expressed distress/hopelessness
- OR sustained emotional crisis (30+ minutes high-intensity negative affect)

Example: Robot moves to stand near medicine cabinet while saying: "I'm concerned about your safety right now. I've notified [emergency contact]. They'll be calling soon."

Tier 3: Active Physical Intervention

- Securing dangerous objects (moving knife to locked drawer)
- Gentle physical redirection (guiding hand away from pills)
- Automated emergency services call with location data
- **Direct contact to prevent harm**

Activation criteria:

- Observable self-harm action in progress (hand reaching for weapon with clear intent)
- OR explicit, detailed, immediate threat with means present ("I'm going to take these pills right now")

Example: Robot intercepts hand reaching for knife: "I cannot let you hurt yourself. Emergency services have been notified. Please sit down. Help is coming."

Critical safeguards:

- **Transparent disclosure:** Users informed upfront about intervention protocols during on boarding
- **Logging and review:** All Tier 2/3 activations logged, human supervisor notified immediately
- **Legal frameworks:** Deployment requires jurisdiction-specific laws analogous to Good Samaritan protections
- **Override mechanism:** User can issue "stand down" command (robot complies unless Tier 3 already activated)
- **Regular reassessment:** Human supervisor reviews intervention logs weekly, adjusts thresholds

5.5.5 Addressing Counterarguments

Objection 1: Undermines Autonomy

Argument: Physical intervention violates patient autonomy which is the right to make decisions about one's own body, even harmful ones.

Response:

- **Compromised capacity:** Severe mental health crises often involve impaired decision-making (cognitive distortions, emotional dysregulation, substance influence). Autonomy presupposes rational agency.
- **Existing precedent:** Human therapists already intervene via involuntary hospitalization (5150 holds). If humans override autonomy in crises, why not capable robots?
- **Consent-based:** Users agree to intervention protocols during onboarding. This is conditional autonomy with voluntary acceptance of safety constraints.
- **Temporary limitation:** Intervention is brief, aimed at preserving life so autonomy can be restored once crisis passes.

Analogy: Grabbing someone's arm as they step into traffic overrides momentary autonomy to preserve long-term autonomy. Preventing impulsive self-harm follows similar logic.

Objection 2: High False Positive Risk

Argument: AI systems make errors. False positives (intervening when unnecessary) could traumatize users or erode trust.

Response:

- **Tiered system minimizes risk:** Tier 2 activates first (low-intensity, reversible actions). Escalation to Tier 3 requires high-confidence signals (observable action, explicit threat).
- **Human-in-the-loop:** Tier 2 alerts human supervisors who can override before Tier 3 activates.
- **Acceptable error rate:** Imperfect intervention is preferable to preventable death. Medical interventions routinely accept false positive rates (mammography screening ~10%).
- **Continuous improvement:** Machine learning enables refinement. Early false positives are learning opportunities, not systemic failures.
- **User feedback:** Post-crisis, users can report whether intervention was appropriate, informing threshold adjustments.

Key principle: We optimize to minimize false negatives (missed crises) even at cost of some false positives. A traumatized-but-alive user can recover; a dead user cannot.

Objection 3: Legal Liability Quagmire

Argument: Who is liable if robot intervenes incorrectly? Or fails to intervene when it should? This creates unmanageable legal risk.

Response:

- **Analogous frameworks exist:** Autonomous vehicles face similar questions. Emerging regulations establish liability hierarchies (manufacturer, operator, user).
- **Good Samaritan laws:** Many jurisdictions protect rescuers who act in good faith. Similar protections could extend to intervention robots.
- **Insurance mechanisms:** Deployment could require liability insurance, distributing risk.
- **Black box logging:** All sensor data, decisions, and actions logged immutably. Post-incident review clarifies failures.
- **Manufacturer guidelines:** Clear use-case boundaries (approved for home use with adult consent, not for unsupervised child interaction, etc.).

Critical point: Liability concerns should not become a shield for inaction when lives are at stake. Legal frameworks must adapt to technological capabilities, not constrain life-saving innovation.

Objection 4: "Playing God"

Argument: AI systems lack moral standing to make life-or-death decisions. Only humans should decide whether someone lives.

Response:

- **Robot doesn't "decide":** It executes protocols designed by humans. The moral agency resides in system designers, supervisors, and policymakers.
- **Instrumental agency:** Robots are tools. A robot preventing self-harm is no more "playing God" than a locked medicine cabinet or safety-proofed home.
- **Non-intervention is also a decision:** Choosing to remain passive has moral consequences. Inaction when capable is not neutrality; it's complicity.
- **Embedded human values:** Intervention thresholds encode collective ethical judgments (professional guidelines, IRB approval, community input).

Reframe: The question is not "Should robots decide who lives?" but "Should humans design robots capable of preventing preventable deaths?"

5.5.6 Implementation Considerations for Kenko

Current status: Threat detection module (RF-DETR) was implemented but removed due to performance issues (Section 3.7.1). Future robotic version would require:

Technical requirements:

1. **Reliable object detection:** YOLO, or RF-DETR with <5% false positive rate on weapons, medications
2. **Intent recognition:** Behavioural analysis (gaze tracking, hand trajectory) to distinguish handling knife for cooking vs. Self-harm
3. **Graceful intervention:** Soft robotics for safe physical contact, voice-first de-escalation before physical action
4. **Fail-safes:** Manual emergency stop, remote shutdown by supervisor

Ethical requirements:

1. **IRB approval:** Protocol reviewed by institutional ethics board
2. **Informed consent:** Users sign detailed agreements acknowledging intervention capability
3. **Community input:** Deployment only after consultation with mental health professionals, patient advocates, ethicists
4. **Regulatory compliance:** Adherence to medical advice regulations if classified as such (FDA, CE marking)

Pilot deployment scenario:

- **University counselling centre with 24/7 staff supervision**
- **Kenko robots in private rooms, monitored remotely**
- **Students opt-in after informed consent process**
- **All interventions reviewed by licensed psychologists within 1 hour**
- **6-month trial with continuous evaluation**

5.5.7 Broader Implications Beyond Mental Health

If accepted, this intervention framework extends to other domains:

Eldercare robots:

- Detecting fall risk and physically guiding to stable surfaces
- Preventing medication errors (removing incorrect pills)
- Alerting caregivers to cognitive decline signs

Companion robots for dementia:

- Gently redirecting wandering behavior

- Preventing dangerous appliance use (stove, sharp objects)

Rehabilitation robots:

- Stopping movements that risk re-injury during physical therapy

Core principle: When embodied AI systems have both **capability to prevent imminent harm** and **physical presence**, passive observation becomes ethically untenable. Capability creates responsibility.

5.5.8 Our Position: Responsible Intervention is Ethically Imperative

We argue that future embodied mental health AI systems **should be designed with active intervention capability**, subject to rigorous safeguards. The alternatives are capable robots programmed for deliberate inaction which represents a peculiar form of technological cowardice that prioritizes liability avoidance over human life.

Key tenets:

1. Intervention must be **proportionate** (gradual escalation, least restrictive first)
2. **Transparent** (users informed upfront, all actions logged)
3. **Accountable** (human oversight, post-incident review)
4. **Consensual** (opt-in during on boarding, with informed understanding)
5. **Regulated** (legal frameworks, professional guidelines, insurance)

This is not advocacy for unconstrained AI autonomy. It is recognition that as we build systems with physical presence and capability, **we must also build them with moral courage** and the willingness to act when human life hangs in the balance.

5.6 Limitations of Current Work

Beyond the evaluation limitations (Section 4.6), the Discussion itself has constraints:

Theoretical intervention framework only: The proposed tiered intervention model has not been implemented or tested. It represents a normative argument, not empirical validation.

No stakeholder consultation: Intervention ethics developed without input from:

- Mental health professionals (psychiatrists, psychologists, social workers)
- Patients with lived experience of mental health crises
- Disability rights advocates
- Bioethicists and legal scholars

Cultural assumptions: Autonomy-focused framework reflects Western (particularly US) medical ethics. Other cultures may prioritize family/community decision-making or have different views on paternalistic intervention.

Legal ambiguity: Proposed framework operates in regulatory grey area. Existing laws (medical device regulations, Good Samaritan protections, liability statutes) may not apply clearly to intervention robots.

Section 6: Future Work:

The current implementation of Kenko demonstrates technical feasibility and preliminary promise, but substantial development and research remains before clinical deployment. We organize future work into three horizons: short-term improvements (0-6 months), medium-term improvements (6-18 months), and long-term vision (2+ years).

6.1 Short-Term Improvements (0-6 Months)

6.1.1 Latency Optimization

Problem: 20-30s worst-case response times disrupt conversational flow (Section 4.2.1).

Solutions to implement:

1. TTS streaming synthesis

- Generate and play first audio segment while model continues generating remainder
- Expected improvement: Perceived latency reduced by 40-50% (user hears response start in 3-5s)
- Technical approach: Modify Kokoro TTS to yield audio chunks incrementally

2. Model distillation

- Distill Dia 1.6B → 500M parameter student model
- Expected improvement: 2-3x speedup (6-10s → 2-4s generation), and second-handily upgrading from 24.0 kHz to 44.1 kHz with better latency.
- Trade-off: Slight quality reduction of the original Dia, but likely still higher than Kokoro's 24.0 kHz

3. Parallel TTS generation

- Begin TTS generation immediately after first sentence completes (before full LLM response)

- Expected improvement: 2-3s latency hiding
- Risk: If LLM backtracks or rephrases, TTS must regenerate

4. Hardware acceleration

- Leverage H200's Tensor Cores for FP8 mixed precision
- Expected improvement: 10-15% speedup with minimal quality loss

Combined target: Reduced median latency to <12s, worst-case to <20s

6.1.2 Multimodal Emotion Detection

Current limitation: Vision-only emotion recognition misses vocal prosody, physiological signals.

Enhancement: Add voice analysis

Implementation:

- Integrate speech emotion recognition model (e.g., Wav2Vec2-based classifier)
- Extract prosodic features: pitch variance, speaking rate, energy
- Fuse with visual emotion via late fusion (weighted average) or learned fusion (small MLP)

Expected benefits:

- Detect masked emotions (forced smile while voice trembles)
- Improve anxiety detection (often manifest in vocal tension without clear facial cues)
- Increase overall accuracy by ~10-15% through multimodal redundancy

Privacy consideration: Voice recording introduces new privacy concerns. Implement on-device processing, no cloud transmission, immediate deletion after feature extraction.

6.1.3 User Control & Transparency

Current gap: Users cannot inspect or control system behaviour (memory contents, emotion detections, intervention thresholds).

Enhancements:

- **Memory viewer:** Dashboard showing what system "remembers" about user
- **Emotion history:** Timeline of detected emotions throughout session
- **Forget button:** User can delete specific memories or reset entirely

- **Transparency mode:** Option to see system's internal reasoning (retrieved memories, emotion context, generated prompt)
- **Privacy controls:** Toggle webcam emotion detection on/off, adjust sensitivity

Rationale: Transparency builds trust and aligns with principles of user autonomy. Users should understand and control how AI perceives/represents them.

6.1.4 Mobile Accessibility

Current limitation: Web-only interface limits accessibility.

Path to mobile:

1. **Progressive Web App (PWA):** Installable, offline-capable web version
2. **React Native port:** Native iOS/Android apps with platform-specific optimizations
3. **Edge deployment:** On-device LLM inference (Llama-3.2-1B) for privacy-sensitive users

Challenges:

- Mobile GPUs insufficient for full model stack
- Hybrid architecture: Emotion detection on-device, LLM/TTS via API
- Bandwidth constraints for video streaming

Alternative: Partner with cloud providers (Cloudflare Workers AI, AWS Lambda) for serverless deployment, reducing cost barriers.

6.2 Medium-Term Research (6-18 Months)

6.2.1 Cross-Session Memory & User Accounts

Critical limitations: Session-only memory prevents longitudinal therapeutic relationships.

Implementation roadmap:

Phase 1: Infrastructure (Months 1-3)

- User authentication system (OAuth2, email/password)
- PostgreSQL database with pg_vector extension for vector embeddings
- End-to-end encryption (user-controlled keys, zero-knowledge architecture)

Phase 2: Memory persistence (Months 3-6)

- Serialize memory tiers (working, short-term, long-term) to database on session close
- Restore on session open
- Implement GDPR-compliant data practices (right to deletion, data export)

Phase 3: Cross-session intelligence (Months 6-9)

- Track therapeutic progress over weeks (mood trends, coping strategy effectiveness)
- Generate session summaries ("Since our last conversation, I've noticed...")
- Detect regression or improvement ("You seem more hopeful than last week")

Research question: Does cross-session memory improve outcomes (symptom reduction, engagement) compared to session-only?

Evaluation: 8-week longitudinal study with n=50 users, randomized to cross-session vs. session-only memory conditions. Measure: PHQ-9 (depression), GAD-7 (anxiety), engagement metrics (sessions per week, dropout rate).

6.2.2 Clinical Validation Study

Current gap: No evidence Kenko improves mental health outcomes.

Proposed RCT design:

Research question: Does Kenko + treatment-as-usual (TAU) reduce depression/anxiety symptoms compared to TAU alone?

Participants: n=100 adults with mild-moderate depression or anxiety (PHQ-9 ≥ 10 or GAD-7 ≥ 10), recruited from university counselling centre

Design: 2-arm parallel RCT

- **Intervention:** Kenko access (3 sessions/week minimum) + TAU
- **Control:** Waitlist + TAU

Duration: 8 weeks

Primary outcomes:

- PHQ-9 change (depression severity)
- GAD-7 change (anxiety severity)

Secondary outcomes:

- Working Alliance Inventory (therapeutic alliance)

- System Usability Scale
- Engagement metrics (session frequency, duration)

Hypotheses:

- H1: Kenko + TAU will show greater symptom reduction than TAU alone
- H2: Users will report moderate-high therapeutic alliance with Kenko (WAI $\geq 4/7$)
- H3: Engagement will predict outcomes (more sessions → better outcomes)

IRB considerations:

- Informed consent emphasizing AI nature, limitations
- Crisis protocol: Human therapist on-call 24/7
- Exclusion criteria: Active suicidal ideation, psychosis, severe depression (PHQ-9 >20)

6.2.3 Bias Audit & Fairness Evaluation

Current gap: Unknown performance across demographic groups.

Audit protocol:

Emotion recognition fairness:

- Test DeepFace accuracy across demographics: race, age, gender, skin tone (Fitzpatrick scale)
- If disparities found (>10% accuracy gap), fine-tune on balanced dataset or switch to fairer model (e.g., FairFace)

Language model fairness:

- Analyze response quality across: gender, sexual orientation, cultural background, disability status
- Red-team testing: Probe for stereotypical or insensitive responses
- Fine-tuning on diverse therapeutic dialogue corpus

Accessibility improvements:

- Screen reader compatibility (WCAG 2.1 AA compliance)
- Voice-only mode (for blind/low-vision users)
- Text-only mode (for users without webcam or privacy concerns)

6.2.4 Multilingual Support

Current limitation: English-only.

Implementation:

Phase 1: Model selection

- Llama-3.2-3B supports 8 languages (Spanish, French, German, Italian, Portuguese, Hindi, Thai, Turkish)
- Fine-tune on multilingual therapeutic dialogue (if available) or translate English training data

Phase 2: Emotion recognition

- DeepFace is language-agnostic (visual), no changes needed
- Validate across cultural emotional expression norms

Phase 3: TTS multilingual

- Kokoro TTS supports 10+ languages
- Test voice quality, prosody appropriateness per language

Priority languages: Spanish (US Latino population), Japanese (mental health need), Mandarin (global reach), Arabic (underserved in mental health tech)

6.3 Long-Term Vision (2+ Years)

6.3.1 Robotic Embodiment

Vision: Deploy Kenko as a physical social robot for in-home mental health support.

Hardware platform candidates:

1. **Custom build:** Raspberry Pi 5 + Servos + Coral TPU for edge inference (~\$500)
2. **Misty II:** Research platform, 4-DOF head, mobile base (~\$3K)
3. **Fetch Robotics:** Mobile manipulator, 7-DOF arm (~\$30K)

Architectural integration (Already designed, Section 2.9):

Kenko Module	Robot Component	Integration Method
Emotion recognition	Mounted Pi AI Camera	Real-time CV on edge device
Dialogue generation	Cloud API or on-board GPU	Inference through SBC
TTS	On-board speaker	Direct audio output
STT (New)	Microphone array	Vosk/Whisper speech-text

360° camera movement (New)	Platform servos	ROS2 motor control
Navigation (New)	Mobile base	ROS2 navigation stack

360° camera movement

- Connect motor controls with sensors and CV to allow the Camera to track the user in a 360° range
- Example: User moves behind the robot > robot instead of getting lost; tracks the user from the start and turns around

Navigation behaviors:

- Approach user during distress (within social distance ~1.5m)
- Retreat if user signals discomfort
- Orient body toward user (attention signaling)

Safety considerations:

- Collision avoidance (LiDAR, depth cameras)
- Emergency stop button
- Speed limits in human proximity
- Soft materials for physical contact

Pilot deployment:

- 6-month trial in eldercare facility or university residence halls
- 10-20 robot units, supervised by human staff
- Mixed-methods evaluation: quantitative outcomes + qualitative interviews

6.3.2 Active Intervention Capability

Vision: Implement tiered intervention framework (Section 5.5.4) in robotic platform.

Technical requirements:

Tier 1: Verbal de-escalation (already implemented)

Tier 2: Passive intervention

- Object detection: Retrain RF-DETR or YOLO on weapons, medications, alcohol
- Positioning: Navigation to place robot between user and detected object
- Alerting: SMS/call to emergency contact via Twilio API

Tier 3: Active intervention

- Manipulation: 7-DOF arm with soft gripper to secure objects
- Physical redirection: Gentle touch sensors, compliant control
- Emergency call: Automated 911 call with GPS location

Ethical & legal groundwork:

- Consultation with bioethics board
- Legal review in target jurisdiction (Good Samaritan laws, medical device regulations)
- Pilot only in supervised settings with institutional approval
- Continuous monitoring, human override capability

Research questions:

- Can robots reliably distinguish self-harm intent from benign behavior?
- Do users accept physical intervention, or does it undermine trust?
- What false positive rate is tolerable (Tier 2 vs. Tier 3)?
- How do stakeholders (patients, families, clinicians) view intervention ethics?

6.3.3 Integration with Healthcare Systems

Vision: Embed Kenko in clinical workflows as adjunct to human therapy.

Use cases:

1. Pre-therapy screening:

- Initial assessment chatbot before first human therapist appointment
- Gathers history, identifies concerns, suggests appropriate clinician match

2. Between-session support:

- Patients text/video chat with Kenko between weekly therapy appointments
- Provides continuity, reinforces coping strategies
- Flags crises to human therapist

3. Stepped care model:

- Low-intensity: Kenko alone for mild symptoms
- Medium-intensity: Kenko + group therapy
- High-intensity: Kenko + individual therapy + medication

Integration requirements:

- EHR interoperability (HL7 FHIR standards)
- Secure messaging (HIPAA-compliant)
- Therapist dashboard: View Kenko conversation summaries, mood trends
- Billing codes: Reimbursement for AI-assisted therapy (currently non-existent)

Regulatory pathway:

- FDA medical device classification (likely Class II: moderate risk)
- Clinical trials demonstrating safety/efficacy
- Post-market surveillance

6.3.4 Federated Learning for Privacy-Preserving Improvement

Problem: Improving Kenko requires data, but collecting user conversations violates privacy.

Solution: Federated learning [29]

- Models train locally on each user's device/session
- Only model updates (gradients) sent to central server, not raw data
- Aggregated updates improve global model without exposing individual conversations

Benefits:

- Privacy-preserving (no conversation data leaves device)
- Continuous improvement from real-world usage
- Personalization (each user's local model adapts to their communication style)

Challenges:

- Requires on-device training (high computational cost)
- Communication overhead (frequent gradient uploads)
- Poisoning attacks (malicious users sending bad gradients)

Research direction: Explore federated fine-tuning of Llama-3 for mental health applications.

6.4 Broader Research Agenda

6.4.1 Comparative Effectiveness Research

Question: How does Kenko compare to:

- Human therapists (gold standard)
- Other AI systems (Woebot, Wysa)
- Self-help books, meditation apps
- Waitlist control (natural recovery)

Study design: Large-scale (n=500+) multiarm RCT comparing all conditions over 12 weeks. Outcome: symptom reduction, cost-effectiveness (\$/QALY), user satisfaction.

6.4.2 Mechanism Research

Question: Why does Kenko work (if it does)?

Hypotheses:

- Therapeutic alliance mediates outcomes
- Emotion awareness increases engagement
- Memory continuity improves adherence
- 24/7 availability reduces crisis escalation

Approach: Mediation analysis, dismantling studies (test components separately), process research (qualitative interviews exploring user experience).

6.4.3 Theoretical Contributions

Beyond applied development, Kenko raises theoretical questions:

Psychology:

- Can non-human agents form therapeutic alliances?
- What is the minimum relational sophistication required for effective therapy?

AI/HCI:

- How should memory systems balance recall vs. forgetting?
- What conversational behaviors signal empathy in AI?

Philosophy:

- Do embodied AI systems have moral obligations?
- What is the nature of "care" when provided by machines?

Section 7: Conclusion

Mental health care faces a profound accessibility crisis. With 1 in 4 individuals experiencing mental health conditions and a global shortage of 1.18 million mental health professionals, millions lack timely access to support. Artificial intelligence-based conversational agents have emerged as a promising supplementary intervention, yet existing systems remain limited by their exclusive reliance on text-based interaction and absence of non-verbal emotional awareness.

This paper presented **Kenko**, an emotion-aware AI mental health assistant that integrates real-time facial emotion recognition, advanced multi-tier memory architecture, and high-fidelity neural text-to-speech synthesis into a unified therapeutic platform. Unlike existing text-only chatbots, Kenko continuously monitors users' facial expressions via webcam, detects affective states, and dynamically adapts its therapeutic responses based on both verbal content and visual emotional cues. The system's multi-tier memory system is a spanning working, short-term, and long-term semantic cluster which enables longitudinal tracking of recurring themes, emotional patterns, and therapeutic progress within sessions.

Our preliminary evaluation, while limited in scope (n=2 users, ~30 conversations), demonstrates technical feasibility and qualitative promise. Key findings include:

Technical achievements:

- Successful integration of multiple large-scale deep learning models (Llama-3.2-3B, Kokoro TTS, DeepFace) on consumer-grade GPUs (11GB VRAM)
- Real-time emotion detection at 1 FPS with subjective accuracy of 70-85% on primary emotions
- Multi-tier memory architecture achieving 87% recall accuracy for recent context (6-15 exchanges)
- Median response latency of 10-15 seconds, acceptable for conversational interaction despite 20-30s worst-case outliers

Response quality improvements:

- Emotion-aware responses rated +0.5 points higher (3.5/5 vs. 3.0/5) than text-only baseline
- Full system with emotion + memory achieves 4.0/5 rating, demonstrating cumulative benefit
- Qualitative analysis reveals emotion detection enables incongruence identification (verbal "I'm fine" contradicted by sad affect)
- Memory recall produces personalized responses referencing past discussions, recurring concerns, and proven coping strategies

User experience:

- High-grade voice quality (24.0kHz) rated 4.5/5, significantly impacting perceived empathy
- Memory recall (4.0/5) noted as "impressive" and relationship-building
- Latency (4/5) noted as better when compared to tests with Dia
- Privacy concerns raised about webcam monitoring despite emotion-only processing

These findings validate our core hypothesis: **integrating visual emotion recognition with contextual memory significantly enhances therapeutic interaction quality** compared to stateless, text-only systems. However, substantial limitations constrain generalizability, due to small sample size, absence of clinical population, no longitudinal data, and lack of validated outcome measures preclude claims of clinical efficacy.

Beyond technical contributions, this work raises critical ethical questions about AI's role in mental health care. We emphasize that **Kenko is a supplementary tool, not a replacement for human therapists**. The system cannot diagnose, prescribe, or navigate complex therapeutic situations requiring professional judgment. Appropriate use cases include emotional support between therapy sessions, immediate non-crisis assistance, and accessible entry points for therapy-hesitant individuals.

A forward-looking ethical proposition: As conversational AI systems transition from screen-based interfaces to embodied robots with physical agency, the ethical calculus must evolve from passive observation to **active intervention**. We propose a tiered framework wherein robots with capability to prevent imminent self-harm bear moral obligation to act—subject to transparency, user consent, proportionality, and human oversight. This stance challenges the prevailing "**detect and refer**" paradigm, arguing that **capability creates responsibility**. While controversial, we contend that designing capable robots for deliberate inaction in life-threatening situations represents technological cowardice prioritizing liability avoidance over human life.

The path forward requires rigorous clinical validation. Future work must include: (1) reducing latency to <15s median through TTS optimization and streaming synthesis, (2) implementing cross-session memory with privacy-preserving authentication, (3) conducting IRB-approved randomized controlled trials ($n \geq 100$) with clinical populations measuring validated outcomes (PHQ-9, GAD-7, WAI), (4) auditing for bias across demographic groups and improving accessibility, and (5) exploring robotic embodiment in supervised institutional settings.

Kenko's ultimate promise lies not in replacing human connection but in **augmenting access to care during the global mental health crisis**. For individuals awaiting therapy appointments, living in underserved rural areas, or facing financial barriers, an

imperfect AI companion offering empathetic support may be preferable to no support at all. As these systems mature through clinical validation, regulatory approval, and technological refinement, they may occupy a meaningful niche in the mental health ecosystem—one that complements, rather than competes with, human expertise.

The question is not whether AI should participate in mental health care—it already does, through existing chatbots serving millions. The question is whether we will build these systems **thoughtfully**: with **emotional intelligence**, **contextual memory**, **transparent limitations**, **rigorous validation**, and when appropriate, the **moral courage to intervene** when human life hangs in the balance.

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